



Two-stage optimization of robotaxi dispatch via recommendation of pick-up/drop-off points and boarding time

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ABSTRACT

As autonomous driving technology reaches a more advanced stage, robotaxis are poised to become a highly promising option for urban mobility. When passengers opt for carpooling, their requested origins/destinations and boarding time (BT) may lead to inefficient robotaxi dispatch. This paper studies a novel robotaxi dispatch problem to recommend pick-up and drop-off (PUDO) points and BT that differ from the requested ones for passengers, considering passenger willingness, i.e., whether to accept these recommendations or not. However, one-stage optimization cannot formulate the realistic decision-making behavior of passengers. To address this issue, a two-stage method is proposed to solve the problem. In the first stage, the PUDO points and BT are optimized and recommended to passengers. In the second stage, with passenger willingness, the routes of robotaxis are planned to optimize travel efficiency. The two stages are formulated as Mixed Integer Programming (MIP) models solved by two algorithms based on Nondominated Sorting Genetic Algorithm II (NSGA-II). The first algorithm is called Nondominated Sorting Genetic Algorithm II via Mass Center (NSGA-MC). It includes a trip similarity determination strategy, a segmented encoding process, and a three-operation evolution to update each fragment of the solution, and introduces a mass center for obtaining a high-quality solution. The second is a Nondominated Sorting Genetic Algorithm II with Improved Evolution (NSGA-IE). It involves an integer encoding process, a new evolution operator for route planning, and a post-iteration selection to choose a final solution from the Pareto-optimal set. The proposed algorithms can balance the conflicting objectives of maximizing the profit per kilometer of robotaxis and minimizing the total travel expense of passengers. The validity of the proposed model is confirmed by an exhaustive search method and a commercial solver CPLEX. The experimental results demonstrate that the proposed algorithms effectively determine feasible solutions and outperform NSGA-II and Multi-objective Evolutionary Algorithm based on Decomposition (MOEA/D) across multiple performance metrics. An ablation study illustrates the benefits of the recommendation of PUDO points and BT. Furthermore, a sensitivity analysis examines the full range of passenger willingness to accept recommendations, ranging from 0 to 100%, with intervals of 25%. Consequently, the proposed method effectively reduces the travel cost for passengers and increases the revenue of robotaxi companies. This work can actively prepare and promote the adoption of

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robotaxis in our real life, thereby contributing to the advancement of intelligent public transportation services.

1. Introduction

Autonomous vehicles integrate various technologies, including computer vision, sensor and communication technology, machine learning, and artificial intelligence (Hussain and Zeadally, 2019). They possess the capability to sense their surroundings and make decisions autonomously, thereby reducing driving accidents and enhancing the convenience and comfort of travel for residents (Eskandarian et al., 2021). In the near future, robotaxis are anticipated to emerge as safe, efficient, and sustainable public transportation options that improve people's lives, potentially revolutionizing urban transportation (Zhang et al., 2024).

Recently, the deployment of high-level autonomous taxi services with human drivers on board has gained popularity worldwide (Dai et al., 2023). Furthermore, some companies, such as Cruise and Waymo, have commenced fully autonomous (no human drivers) testing (Li, 2023; Carlson, 2022). Robotaxis differ from the current human-driven taxis (HDTs) in the following aspects (Narayanan et al., 2020; Dandl and Bogenberger, 2019; Zeng et al., 2022): 1) The elimination of human drivers enables robotaxis to substantially reduce labor cost. Moreover, the travel cost becomes primarily distance-based and time-based rather than constrained by driver availability, and a centrally managed robotaxi fleet can provide 24/7 dispatching services (Tavor and Raviv, 2023); 2) The number of available robotaxis for dispatch can be dynamically adjusted according to the number of requests, whereas the number of HDTs is typically fixed. A case study demonstrates that switching from HDTs to robotaxis can potentially reduce the fleet size by 59% while maintaining the quality of service (Lokhandwala and Cai, 2018); and 3) In major ride-hailing platforms such as Uber, Lyft, and DiDi (Hu and Dong, 2022; Ao et al., 2024), an HDT driver typically prioritizes individual earnings, potentially impeding the maximization of profit per kilometer for all HDTs. This work addresses a novel robotaxi dispatch problem, encouraging passengers to accept recommended pick-up and drop-off (PUDO) points that may differ from their requested origins and destinations and to take a robotaxi earlier or later than their scheduled time.

While most studies focus on carpooling, few simultaneously optimize passengers' PUDO points and boarding time (BT), a critical oversight that degrades passenger experience and increases the travel cost of robotaxis. Although some studies account for changes to PUDO points, they do not consider passenger willingness to accept the modifications (termed "passenger willingness" in the following). Furthermore, from the passenger's perspective, an optimal pricing policy should maximize social welfare, e.g., reducing their travel expense (Wang et al., 2016). However, most studies calculate travel cost solely based on distance, with limited efforts to design suitable pricing policies for the changes of PUDO points and BT. Hence, this paper proposes a two-stage robotaxi dispatch optimization that incorporates all the aforementioned factors. Notably, one-stage optimization fails to generate specific recommendations of PUDO points and BT, which cannot reflect the realistic decision-making behavior of passengers (Li et al., 2025). The novel contributions of this study are as follows:

- 1) A novel two-stage robotaxi dispatch problem: A two-stage robotaxi dispatch via the recommendation of PUDO points and BT (RD-RPT) is studied, which changes both PUDO points and BT of passengers and considers passenger willingness (i.e., whether the passengers accept the recommendations or not).
- 2) Multi-objective optimization models: Two multi-objective optimization models (Model 1 and Model 2) are formulated to solve the robotaxi dispatch problem. They have different decision variables, and Model 2 is conditional on Model 1's outputs. These models aim to minimize the total travel expense of passengers and maximize the profit per kilometer of robotaxis. A pricing policy is introduced to incentivize passengers to accept the recommendation by offering discounts, while simultaneously using it to calculate the associated cost and profit.
- 3) A two-stage method: It involves two modified metaheuristic algorithms: Nondominated Sorting Genetic Algorithm II via Mass Center (NSGA-MC) and Nondominated Sorting Genetic Algorithm II with Improved Evolution (NSGA-IE). In NSGA-MC, a trip similarity determination strategy is developed to assess the probability of carpooling among requests; a mass center (MC) of the initial population is calculated to select high-quality solutions on two objectives; and a three-operation evolution is presented in which each individual is first decomposed into three segments, updated independently, and then recombined into a complete individual. In NSGA-IE, an improved evolution operator is developed to optimize routes of robotaxis; and a post-iteration selection is designed to choose the best solution in the Pareto-optimal set.

In addition, the validity of the proposed model is validated by an exhaustive search method and CPLEX. The proposed algorithms are compared with two classical optimization algorithms: Nondominated Sorting Genetic Algorithm II (NSGA-II) (Deb et al., 2002) and Multi-objective Evolutionary Algorithm based on Decomposition (MOEA/D) (Zhang and Li, 2007) via experiments. The results demonstrate the superiority of the proposed algorithm in solving the problem. An ablation study is conducted to determine the performance of our method. The experiments cover the full range of the percentage of passengers who accept recommendations, from 0 to 100%, thereby analyzing the impact of passenger willingness on results.

The rest of this paper is organized as follows. Section 2 provides an overview of related work. Section 3 proposes mathematical models for a two-stage robotaxi dispatch problem. Section 4 elaborates on an algorithm to solve the problem. Section 5 presents and analyzes the experimental results. Finally, Section 6 concludes the study and discusses future work.

2. Related work

This section presents a literature review related to the studied problem, including general taxi dispatch problems and carpooling service problems. Among them, the research considering PUDO points or BT and their recommendations in robotaxi dispatch is also reviewed.

2.1. Taxi dispatch problem

An efficient HDT dispatch model aims to provide point-to-point high-quality travel services for urban residents, which needs to solve a taxi-passenger matching problem and a vehicle routing problem (VRP) (Mańdziuk, 2019; Seo and Asakura, 2022). The drivers of HDTs primarily rely on personal experience or mobile platforms to search for passengers on the street, aiming to maximize private profit. However, such self-interested and uncoordinated behaviors of drivers often lead to spatiotemporal imbalance between taxi supply and passenger demand (Chen et al., 2021). To mitigate this issue, various mathematical programming methods are proposed to maximize the number of passengers served and minimize the travel cost of taxis. For instance, Miao et al. (2019) develop a data-driven robust taxi dispatch framework, solving it using the minimax theorem and strong duality, thereby reducing the demand–supply ratio error and the total empty-running distance of taxis. Kim et al. (2020, 2022) transform a multi-objective taxi dispatch problem into a network flow problem, employing a minimum cost maximum flow algorithm to fulfill as many ride requests as possible while minimizing vehicle relocation cost. Duan et al. (2020) introduce a dispatch strategy employing a convex optimization solver in a hybrid mode involving immediate and reserved requests.

Some heuristic and metaheuristic algorithms are used to solve the taxi dispatch problem (Chen et al., 2022). Zhu et al. (2019) propose a path-planning strategy based on a greedy algorithm to serve enough customers with the minimum vehicle cost. Gong et al. (2021) employ a differential evolution algorithm to optimize the taxi-passenger matching, as the greedy algorithm may restrict system-wide profitability. In recent years, there have been a few studies on robotaxi dispatch based on deep reinforcement learning (Si et al., 2024). For example, Rivière and Chung (2022) introduce a deep reinforcement learning approach to minimize the expected passenger waiting time. Liu et al. (2022) integrate rich contexts into deep reinforcement learning to enable efficient taxi reallocations. However, these models require retraining if scenarios change, entailing significant convergence time.

In the field of taxi dispatch, the vehicle routing problem (VRP), a critical factor in improving operational efficiency, is evolving from the traditional route optimization to incorporate vehicle capacity, energy consumption, and travel cost. Feng et al. (2021) present a capacitated VRP and develop a specialized evolutionary multitasking algorithm to minimize travel cost. Xu et al. (2025) design an adaptive hyper-heuristic algorithm based on the genetic algorithm (GA) to reduce the distribution cost in a community group-buying VRP. Wang et al. (2024) introduce a multidepot VRP and incorporate GA and particle swarm optimization (PSO) to minimize the operating cost and maximize the total profit. To reduce carbon emissions, Guo et al. (2018) propose a dynamic VRP model with hard-time windows. Abdulaal et al. (2017) address a multivariant VRP by designing a modified custom genetic algorithm integrated with an embedded Markov decision process for electric vehicles. While existing studies on green logistics (Wang et al., 2018; Wang et al., 2020) and VRP (Wang et al., 2022; Wang et al., 2023) offer a significant contribution, challenges remain in addressing high passenger demands and efficient taxi-passenger matching in VRP-based taxi dispatch for future research.

NSGA-II is a well-established algorithm for multi-objective optimization aiming to achieve Pareto optimality (Deb et al., 2002). Numerous variants have been proposed to address transportation-related problems, including taxi dispatch problems. For example, Pourvaziri et al. (2024) integrate neural networks into the evolutionary process of NSGA-II to plan charging stations. Amirsahami et al. (2025) and Shi et al. (2026) enhance NSGA-II by incorporating neighborhood search for a truck-drone collaborative relief chain and a dynamic scheduling of emergency supplies, respectively. Qi et al. (2025) design a selection strategy via pandemic degrees in NSGA-II to obtain an adaptive taxi dispatch plan in the pandemic era. These studies improve NSGA-II based on the characteristics of their studied optimization problems, which provide a methodological guideline for combinatorial optimization problems in transportation.

Table 1

Factors considered in both prior studies and our work in taxi dispatch.

Reference	Carpooling	Specific PUDO-point recommendation	Specific boarding-time recommendation	Passenger willingness	Pricing
Kim et al., 2020, 2022; and Duan et al., 2020.	✗	✗	✗	–	✗
Santi et al., 2014; Luo et al., 2023; Huang et al., 2019; Chau et al., 2022; and Li et al., 2021.	✓	✗	✗	–	✗
Lu et al., 2024.	✓	✗	✗	–	✓
Fielbaum et al., 2021; Zhu et al., 2022; Yao et al., 2021; Park et al., 2023; Lotze et al., 2022; Fielbaum, 2022; Zhu et al., 2021; Nishida et al., 2022; Gökay et al., 2019; and Li et al., 2018.	✓	✓	✗	✗	✗
Cipolina-Kun et al., 2023.	✓	✓	✗	✗	✓
Ding et al., 2024.	✗	✓	✗	✗	✗
Li et al., 2025.	✓	✗	✗	✓	✓
Our work	✓	✓	✓	✓	✓

● Note: The symbol “–” represents that the factor does not apply to particular references.

However, the existing variants of NSGA-II are tailored to their specific scenarios only. Hence, novel variants are developed to address the studied robotaxi dispatch problem.

2.2. Carpooling service problem

In a taxi dispatch problem, shared on-demand mobility systems are commonly referred to as “carpooling” or “ridesharing” systems, where different passengers with somewhat similar routes share a vehicle (Santi et al., 2014; Fielbaum et al., 2021). Carpooling modes help alleviate traffic congestion and effectively reduce the economic pressure on passengers (Luo et al., 2023). Several studies formulate optimization models for carpool service problems. Huang et al. (2019) propose an ant path-oriented carpooling allocation approach to maximize the number of matched passengers and seat usage rates. Chau et al. (2022) introduce a fair cost-sharing mechanism to equitably distribute travel cost fairly among passengers. Li et al. (2021) present a data-driven robust optimization approach by integrating gated recurrent unit predictions with a one-stage robust optimization model to address carpool service problems under travel time uncertainty. Lu et al. (2024) develop a utility-based compensation pricing method to mitigate inequity issues in carpooling arising from variations in waiting and detour time. Qi et al. (2025) focus on carpooling in the pandemic era and develop a low-contact robotaxi dispatch method. However, carpooling induces extra travel time for the passengers in terms of delay because vehicles may need to make long detours to accommodate all passengers’ origins and destinations (Cáp and Alonso-Mora, 2018).

If passengers opt for PUDO points close to their initial origins and destinations, vehicle detours may be reduced. Using real data analysis, Ding et al. (2024) conclude that repositioning the PUDO points within a short walking distance can enhance the potential efficiency and environmental benefits. In recent years, some literature has studied taxi dispatch problems considering changing PUDO points (Fielbaum et al., 2021; Zhu et al., 2022; Yao et al., 2021; Park et al., 2023; Lotze et al., 2022), which may require passengers to walk to specific points. For instance, Fielbaum (2022) optimizes the route of a single vehicle and selects precise PUDO points for passengers. Zhu et al. (2021) design a demand-side cooperative shared automated mobility service framework and address optimal ride-matching problems for four vehicles. Other studies require passengers to walk to a designated meeting point (Nishida et al., 2022; Gökyay et al., 2019), which is a gathering location of passengers from multiple requests. Li et al. (2018) introduce meeting points to a ridesharing system and model it as a mixed integer linear program. Based on game-theoretic principles, Cipolina-Kun et al. (2023) propose a method for determining optimal meeting points, incorporating an equitable cost allocation approach. Additionally, the optimization of PUDO points is closely related to on-demand services. In addition to taxi dispatch, customized on-demand bus services optimize bus routes and determine passenger PUDO points (Zhen et al., 2023); and truck-drone collaborative delivery systems optimize drone launching and landing locations for each request (Zhen et al., 2025), which lays a solid methodological foundation for further optimization of PUDO points in robotaxi dispatch. However, the aforementioned studies seldom consider passengers’ willingness to walk a specific distance to a recommended PUDO point.

In addition to the PUDO points, the BT of passengers is a crucial factor in a robotaxi dispatch system. It is often treated as a constraint in prior studies. Huang et al. (2019, 2020) set a time window to restrict the BT of passengers and use an evolutionary algorithm and an ant colony optimizer to solve it. Guo et al. (2021) improve decision rules for dynamic timeframe segmentation to gather as many passengers as possible in a timeframe. Acknowledging the influence of passenger distribution and traffic flow on the arrival time of robotaxis, Liang et al. (2020) consider dynamic travel time, delving into a dial-a-ride problem in robotaxi scenarios. Zhang et al. (2022) employ a co-prediction method to assist drivers in quickly finding passengers and reducing the waiting time of passengers. Recommending a suitable BT within an acceptable timeframe can optimize taxi-passenger matching, thereby enhancing the service quality of taxis and passenger satisfaction. Although Li et al. (2025) optimize the BT of passengers, they are unable to provide passengers with specific adjustments to BT. A comparative analysis of the factors considered in related studies and the present study is presented in Table 1. Hence, this paper proposes a two-stage robotaxi dispatch method that incorporates all the aforementioned factors and provides passengers with specific recommendations regarding PUDO points and BT.

3. Problem description and formulation

This section introduces a two-stage robotaxi dispatch via the recommendation of PUDO points and BT (RD-RPT), formulated as a multi-objective mixed integer non-linear programming model. Abbreviations used are listed in Table 2. Let $\mathbb{R}$ represent a set of real numbers, $\mathbb{N}=\{0, 1, 2, \dots\}$ denote a set of all natural numbers, $\mathbb{N}^+=\mathbb{N}\setminus\{0\}$, $\mathbb{N}_m=\{0, 1, 2, \dots, m\}$, and $\mathbb{N}_m^+=\mathbb{N}_m\setminus\{0\}$. Specific notations are defined in Table 3.

3.1. Problem description

A request refers to a passenger’s demand for transportation from an origin to a destination and can be categorized into two types: immediate requests and reserved requests. Immediate requests necessitate instant service, whereas reserved ones specify a future time for delivery. Reserved requests are preferable as they allow sufficient time for robotaxi dispatch. This work focuses solely on reserved requests. The sets of robotaxis and requests are denoted by $V=\{1, \dots, i, \dots, m\}$ and $R=\{1, \dots, j, \dots, n\}$, respectively, where m and n denote the number of robotaxis and requests, respectively. A tuple (s_i, q_i) represents robotaxi i , where s_i represents its initial location, and q_i denotes its seat capacity, i.e., the maximum number of passengers that can be carried by robotaxi i ; A quadruple (o_j, d_j, t_j, η_j) represents request j , where o_j , d_j , and t_j indicate its origin, destination, and BT, respectively, and η_j represents the number of passengers.

Table 2

A list of abbreviations.

Abbreviations	Full name
PU	Pick-up
DO	Drop-off
PUDO	Pick-up and drop-off
BT	Boarding time
RD-RPT	Robotaxi dispatch via the recommendation of PUDO points and boarding time
MOEA/D	Multi-objective Evolutionary Algorithm based on Decomposition
NSGA-II	Nondominated Sorting Genetic Algorithm II
NSGA-MC	Nondominated Sorting Genetic Algorithm II via Mass Center
NSGA-IE	Nondominated Sorting Genetic Algorithm II with Improved Evolution
RD-NR	Robotaxi dispatch without recommendation
RD-RP	Robotaxi dispatch via the recommendation of PUDO points
RD-RT	Robotaxi dispatch via the recommendation of BT

Table 3

Notations used in the model.

Notation	Definition
<i>Sets</i>	
V	Set of robotaxis.
R	Set of requests.
$\mathbb{O}_j$	Set of candidate PU points for request $j, j \in R$.
$\mathbb{D}_j$	Set of candidate DO points for request $j, j \in R$.
$\mathcal{N}$	Set of nodes, including all initial locations of robotaxis and origins and destinations of requests.
$\mathcal{A}$	Set of arcs in the road network.
<i>Parameters</i>	
s_i	Initial location of robotaxi $i, i \in V$.
q_i	Seat capacity of robotaxi $i, i \in V$.
o_j	Origin of request $j, j \in R$.
d_j	Destination of request $j, j \in R$.
t_j	Initial BT of request $j, j \in R$.
η_j	Number of passengers in request $j, j \in R$.
φ_j	Passenger willingness to accept the recommended BT: $\varphi_j = 1$ if passengers in request j accept the recommendation of BT; otherwise, $\varphi_j = 0, j \in R$.
ψ_j	Passenger willingness to accept walking to the recommended point: $\psi_j = 1$ if passengers in request j accept the recommendation of PUDO points; otherwise, $\psi_j = 0, j \in R$.
o_j^h	Candidate PU point in $\mathbb{O}_j, h \in \{1, 2, \dots, \mathbb{O}_j \}$.
d_j^h	Candidate DO point in $\mathbb{D}_j, h \in \{1, 2, \dots, \mathbb{D}_j \}$.
$l(u, v)$	Robotaxi's driving distance from node u to v in a road network, $u, v \in \mathcal{N}$.
$\ell(u, v)$	Passenger's walking distance between nodes u and $v, u, v \in \mathcal{N}$.
$\bar{c}$	Cost per kilometer of robotaxis.
<i>Decision variables</i>	
$x_{u,v}^i$	Binary variable that equals 1 if robotaxi i passes arc (u, v) , and otherwise equals 0.
$y_{u,v}^{ij}$	Binary variable that equals 1 if robotaxi i carries passengers in request j along arc (u, v) , and otherwise equals 0.
$\hat{a}_{jh}$	Binary variable that equals 1 if passengers in request j are picked up at o_j^h , and otherwise equals 0, where $j \in R$ and $h \in \{1, 2, \dots, \mathbb{O}_j \}$.
$\hat{a}_{jh}$	Binary variable that equals 1 if passengers in request j are dropped off at d_j^h , and otherwise equals 0, where $j \in R$ and $h \in \{1, 2, \dots, \mathbb{D}_j \}$.
u_j	Boarding time (BT) offset, $u_j \in R$. If $u_j < 0$, the BT is $ u_j $ minutes earlier; otherwise, the BT is $ u_j $ minutes later, where $j \in R$.

A set of candidate pick-up (PU) points is denoted as $\mathbb{O}_j$, where $o_j^h \in \mathbb{O}_j$ denotes the h -th potential point to be recommended to request j , and $|\mathbb{O}_j|$ is the number of points in it. For example, $\mathbb{O}_1 = \{o_1^1, o_1^2\}$ is for r_1 and $|\mathbb{O}_1| = 2$. Similarly, the set of candidate drop-off (DO) points is denoted as $\mathbb{D}_j$, where $d_j^h \in \mathbb{D}_j$ denotes the h -th point. The robotaxis are dispatched based on a directed graph $\mathcal{G}(\mathcal{N}, \mathcal{A})$, where

$$\mathcal{N} = \left(\bigcup_{i \in V} \{s_i\} \right) \cup \left(\bigcup_{j \in R} \{o_j, d_j\} \right) \cup \left(\bigcup_{j \in R} \mathbb{O}_j \right) \cup \left(\bigcup_{j \in R} \mathbb{D}_j \right)$$

represents a set of all nodes, including all initial locations of robotaxis, origins and destinations of requests, and candidate PUDO points, and $\mathcal{A}$ represents a set of arcs connecting any pair of nodes.

Now, the problem is illustrated with an example. Suppose there are three robotaxis 1–3 and three requests 1–3, as depicted in Fig. 1. A road network is discretized into a grid with five longitudinal and four lateral lanes, colored in gray, where the arrows of lanes indicate the direction of traffic flow. In (a), the black lines represent initial routes of robotaxis 1–3, $t_1 = t_2$, and requests 1 and 2 are served by robotaxis 1 and 2, respectively. Due to the one-way road network, robotaxis 3 must detour to pick up passengers in request 3. If t_2 is adjusted to 9:03 am, and o_3 is modified by directing passengers in request 3 to walk to a nearby intersection to board robotaxi 3, passengers in requests 1 and 2 can carpool and be served exclusively by robotaxi 1. This adjustment conserves a robotaxi for other

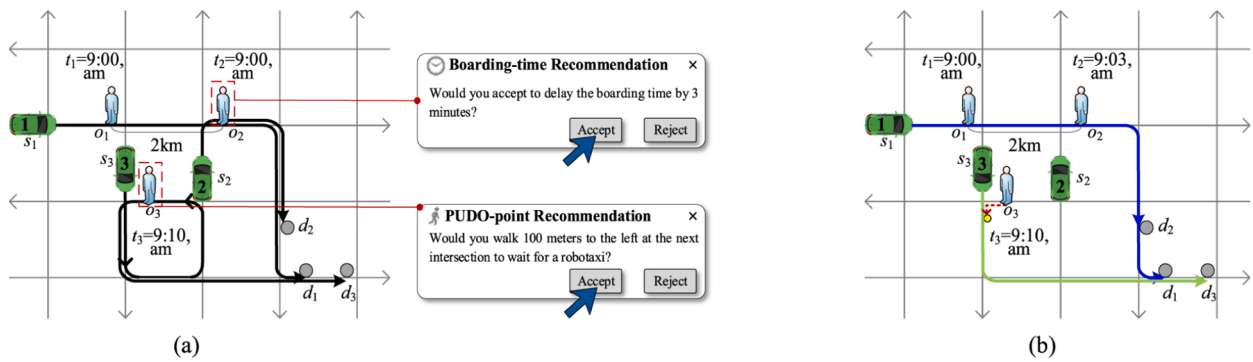


Fig. 1. An example of recommending PUDO points and BT: (a) A Robotaxi dispatch plan without recommendation; and (b) A robotaxi dispatch plan with the recommendation of PUDO points and boarding time.

services and allows robotaxi 3 to avoid a detour. Subsequently, passengers in requests 2 and 3 are prompted to indicate their willingness to accept the updated PU point or revised BT. In (b), a red dashed line denotes the walking route of the passenger in request 3. Based on passenger willingness, new routes of robotaxis are generated, where the blue and green lines represent the routes of robotaxis 1 and 3, respectively. Such modifications yield evident social and economic benefits, i.e., fewer robotaxis with reduced driving distance can fulfill the travel demands at lower cost to passengers.

The framework of the proposed method is illustrated in Fig. 2. Each robotaxi is dispatched by the control center and cannot plan its route on its own. Its behavior helps achieve global goals and is not driven by self-interest. In the first stage, passengers send their requests to the control center. Subsequently, the control center utilizes NSGA-MC to generate new PUDO points and BT. Subsequently, passengers are queried about whether they wish to modify their initial origin, destination, and BT. In the second stage, the passengers' responses are relayed back to the control center. Finally, the match between robotaxis and requests is completed, and the driving routes for the robotaxis are planned via NSGA-IE.

3.2. Mathematical models

To address the two-stage robotaxi dispatch problem, two mathematical models are formulated: Model 1 and Model 2. They have different decision variables, and the second stage is conditional on the first-stage outputs. Although the two models are designed to address the same optimization objective, they differ in their decision spaces, as they correspond to different parts of the overall problem-solving procedure. Specifically, Model 1 corresponds to the first stage, which determines the PUDO points, BT offset, and the preliminary routes of robotaxis (not final routes). Model 2, building upon the optimized PUDO points and BT obtained from Model 1, generates the final routes of robotaxis based on the passenger willingness. For clarity, a general model is presented first, and then the specific variables and formulas associated with Model 1 and Model 2 are introduced, respectively.

Decision variables are defined in Table 3. $x_{u,v}^i \in \{0, 1\}$ is a binary variable, where $x_{u,v}^i = 1$ if robotaxi i travels from node u to node v ,

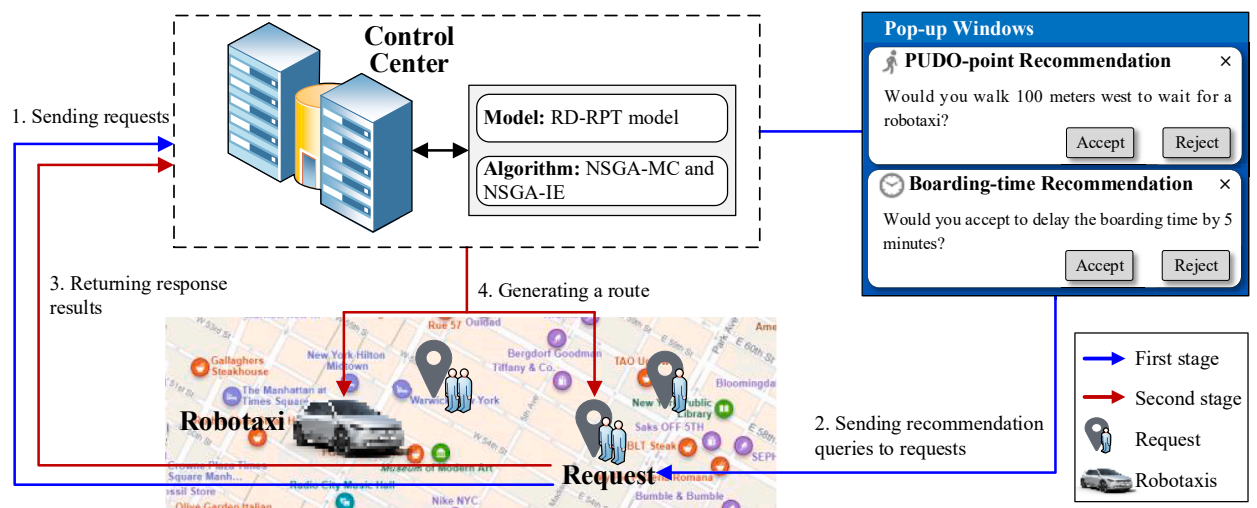


Fig. 2. Framework of the method to solve RD-RPT.

and otherwise $x_{u,v}^i=0$; and $y_{u,v}^{ij} \in \{0, 1\}$ is a binary variable, where $y_{u,v}^{ij}=1$ if robotaxi i carries passengers in request j along the arc (u, v) , and otherwise $y_{u,v}^{ij}=0$. The variables $\check{a}_{jh}$ and $\hat{a}_{jh} \in \{0, 1\}$ indicate whether the candidate PUDO points are recommended, respectively. Specifically, $\check{a}_{jh}=1$ if passengers in request j are picked up at o_j^h ; otherwise, $\check{a}_{jh}=0$. Similarly, $\hat{a}_{jh}=1$ if passengers in request j are dropped off at d_j^h ; otherwise, $\hat{a}_{jh}=0$. The variable u_j denotes a BT offset, representing the change in BT. If $u_j < 0$, the BT is $-u_j$ minutes earlier; otherwise, the BT is u_j minutes later. Subsequently, a general model is formulated as follows:

The recommended PUDO points of request j are denoted by $\bar{o}_j$ and $\bar{d}_j$, respectively.

$$\bar{o}_j = \begin{cases} o_j^h, & \text{if } \check{a}_{jh} = 1 \\ o_j, & \text{otherwise} \end{cases}, o_j^h \in \mathbb{O}_j \quad (1)$$

$$\bar{d}_j = \begin{cases} d_j^h, & \text{if } \hat{a}_{jh} = 1 \\ d_j, & \text{otherwise} \end{cases}, d_j^h \in \mathbb{D}_j \quad (2)$$

The travel cost of a robotaxi is proportional to its travel distance. Here, the cost per kilometer of the robotaxi is denoted by $\bar{c}$. Note that energy consumption, depreciation, and maintenance cost are all allocated to each kilometer traveled by robotaxis (Ren et al., 2023). The labor cost is excluded, as robotaxis are driverless. Hence, the total travel cost of the robotaxi is defined as:

$$c_i = \bar{c} \sum_{(u,v) \in \mathcal{A}} x_{u,v}^i l(u, v) \quad (3)$$

where $l(u, v)$ is the driving distance between two adjacent nodes.

Without the recommendation of PUDO points, the shortest driving distance for a robotaxi to travel from o_j to d_j is denoted as $l(o_j, d_j)$, abbreviated as l_j . Travel expense is the price paid by passengers for a request, which correlates positively with the shortest distance from their origin to their destination. Let a robotaxi base fare be e_0 , and passengers pay c_r per kilometer (Li et al., 2018). Hence, the travel expense of passengers in request j is defined as:

$$\hat{e}_j = e_0 + c_r l_j \quad (4)$$

Carpooling of two or more requests may increase the driving distance of a robotaxi due to a detour, potentially reducing passenger satisfaction. Here, a detour rate is defined as (Zhu et al., 2019):

$$\delta_j = \frac{\sum_{(u,v) \in \mathcal{A}} y_{u,v}^{ij} l(u, v) - l_j}{l_j} \quad (5)$$

Passengers may experience a detour caused by carpooling or a reduction in actual driving distance due to changes in PUDO points, i.e., a negative detour. The new driving distance may be less than l_j according to (5) because of the recommendation of PUDO points, thereby resulting in a negative detour rate. Hence, $\gamma_j = \gamma_0$ if $\delta_j \leq 0$, where γ_0 is a constant representing a carpooling discount. The discount for detours is defined as:

$$\gamma_j = \begin{cases} \alpha \delta_j, & \text{if } \delta_j > 0 \\ \gamma_0, & \text{otherwise} \end{cases} \quad (6)$$

where α is a parameter relating to detours.

Whether passengers in request j are willing to accept the recommendation to walk to recommended PUDO points is denoted by ψ_j : $\psi_j=1$ if passengers accept it; otherwise, $\psi_j=0$. The walking distance of passengers in request j is the sum of the walking distance from o_j to $\bar{o}_j$ denoted by $\ell(o_j, \bar{o}_j)$ and that from $\bar{d}_j$ to d_j denoted by $\ell(\bar{d}_j, d_j)$:

$$\hat{l}_j = \psi_j \left(\sum_{h=1}^{|\mathbb{O}_j|} \check{a}_{jh} \ell(o_j, o_j^h) + \sum_{h=1}^{|\mathbb{D}_j|} \hat{a}_{jh} \ell(d_j^h, d_j) \right) \quad (7)$$

which can be simplified as $\hat{l}_j = \psi_j (\ell(o_j, \bar{o}_j) + \ell(\bar{d}_j, d_j))$.

To compensate for the reduction in passenger satisfaction caused by walking, a financial discount is provided, defined as:

$$\varpi_j = \beta \hat{l}_j \quad (8)$$

where β is a parameter.

Additionally, changing BT also warrants a discount for passengers. The discount for changing BT, denoted by ε_j , is calculated by:

$$\varepsilon_j = \frac{\lambda \varphi_j |u_j|}{u^+ - u^-} \quad (9)$$

where λ is a parameter; and the passenger willingness for time-changing, i.e., being picked up earlier/later or not, is denoted by φ_j :

$\varphi_j=1$ if passengers accept it; otherwise, $\varphi_j=0$.

From the above formula, the total discount $1 - \gamma_j - \varpi_j - \varepsilon_j$ is obtained according to (6), (8), and (9). Hence, the travel expense of request j is denoted by e_j , defined as:

$$e_j = (1 - \gamma_j - \varpi_j - \varepsilon_j) \hat{e}_j \quad (10)$$

The total travel expense of all passengers is calculated by:

$$f_1 = \sum_{j=1}^n e_j \quad (11)$$

The revenue of robotaxis equals the total travel expense of passengers. Therefore, the total profit of all robotaxis is defined as subtracting the total travel cost from the total revenue, denoted by:

$$\hat{p} = \sum_{j=1}^n e_j - \sum_{i=1}^m c_i \quad (12)$$

The profit per kilometer is defined as the total profit divided by the total driving distance of robotaxis, denoted by:

$$\begin{aligned} f_2 &= \hat{p} \left(\sum_{i=1}^m \sum_{(u,v) \in \mathcal{A}} x_{u,v}^i l(u,v) \right)^{-1} \\ &= \frac{\sum_{j=1}^n e_j - \sum_{i=1}^m c_i}{\sum_{i=1}^m \sum_{(u,v) \in \mathcal{A}} x_{u,v}^i l(u,v)} \\ &= \frac{\sum_{j=1}^n e_j - \sum_{i=1}^m \sum_{(u,v) \in \mathcal{A}} \bar{c} x_{u,v}^i l(u,v)}{\sum_{i=1}^m \sum_{(u,v) \in \mathcal{A}} x_{u,v}^i l(u,v)} \\ &= \frac{\sum_{j=1}^n e_j}{\sum_{i=1}^m \sum_{(u,v) \in \mathcal{A}} x_{u,v}^i l(u,v)} - \bar{c} \end{aligned} \quad (13)$$

Objective function (14) represents the expected minimum total travel expense of passengers. This model reduces the travel expense by offering passengers a discount through carpooling and recommending different BT and PUDO points. Such discounts encourage passengers to continue to choose robotaxis for their future trips. However, reducing the travel expense of passengers implies reducing the total revenue of robotaxis. Thus, objective function (15) is utilized to maximize the expected profit per kilometer of robotaxis. The profit of robotaxis is influenced not only by the travel expense of passengers but also by the total travel distance of robotaxis. Consequently, the travel expense may not be negatively correlated with the profit of robotaxis. This demonstrates that the two objective functions are not inherently conflicting and can coexist.

$$\min f_1 \quad (14)$$

$$\max f_2 \quad (15)$$

s.t.

$$\sum_{i=1}^m \sum_{v \in \mathcal{I} \setminus \{u\}} x_{u,v}^i \leq 1, \forall u \in \mathcal{A} \quad (16)$$

$$x_{v,s_i}^i = 0, \forall i \in V, \forall v \in \mathcal{A} \quad (17)$$

$$\sum_{i=1}^m \sum_{v \in \mathcal{I} \setminus \{\bar{o}_j\}} x_{v,\bar{o}_j}^i = \sum_{i=1}^m \sum_{v \in \mathcal{I} \setminus \{\bar{d}_j\}} x_{v,\bar{d}_j}^i = 1, \forall j \in R \quad (18)$$

$$\sum_{v \in \mathcal{I} \setminus \{\bar{o}_j\}} x_{v,\bar{o}_j}^i - \sum_{v \in \mathcal{I} \setminus \{\bar{d}_j\}} x_{v,\bar{d}_j}^i = 0, \forall i \in V, \forall j \in R \quad (19)$$

$$\sum_{i=1}^m \sum_{v: (\bar{o}_j, v) \in \mathcal{A}} y_{\bar{o}_j, v}^{ij} = 1, \forall j \in R \quad (20)$$

$$\sum_{i=1}^m \sum_{v: (v, \bar{d}_j) \in \mathcal{A}} y_{v, \bar{d}_j}^{ij} = 1, \forall j \in R \quad (21)$$

$$\sum_{v \in \mathcal{V} \setminus \{u\}} y_{v,u}^{ij} - \sum_{v' \in \mathcal{V} \setminus \{u\}} y_{u,v'}^{ij} = 0, \forall i \in V, \forall j \in R, \forall u \in \mathcal{V} \setminus \left(\left(\bigcup_{i \in V} \{s_i\} \right) \cup \{o_j, d_j\} \cup \mathbb{O}_j \cup \mathbb{D}_j \right) \quad (22)$$

$$y_{u,v}^{ij} \leq x_{u,v}^i, \forall i \in V, \forall j \in R, \forall (u, v) \in \mathcal{A} \quad (23)$$

$$\sum_{v \in \mathcal{V} \setminus \{\bar{o}_j\}} y_{v,\bar{o}_j}^{ij} = 0, \forall i \in V, \forall j \in R \quad (24)$$

$$\sum_{v \in \mathcal{V} \setminus \{\bar{d}_j\}} y_{\bar{d}_j,v}^{ij} = 0, \forall i \in V, \forall j \in R \quad (25)$$

$$\sum_{h=1}^{|\mathbb{O}_j|} \check{a}_{jh} \leq 1, \forall j \in R \quad (26)$$

$$\sum_{h=1}^{|\mathbb{D}_j|} \hat{a}_{jh} \leq 1, \forall j \in R \quad (27)$$

$$\sum_{h=1}^{|\mathbb{O}_j|} \check{a}_{jh} \ell(o_j, o_j^h) \leq w_{max}, \forall j \in R \quad (28)$$

$$\sum_{h=1}^{|\mathbb{D}_j|} \hat{a}_{jh} \ell(d_j^h, d_j) \leq w_{max}, \forall j \in R \quad (29)$$

$$u^- \leq u_j \leq u^+, \forall j \in R \quad (30)$$

$$\delta_j \leq \Delta, \forall j \in R \quad (31)$$

$$t_0 + \frac{\sum_{i=1}^m \sum_{v \in \mathcal{V}} x_{v,\bar{o}_j}^i l(v, \bar{o}_j)}{\bar{v}} \leq t_j + \varphi_j u_j, \forall j \in R \quad (32)$$

$$\sum_{j=1}^n \eta_j y_{u,v}^{ij} \leq q_i, \forall i \in V, \forall (u, v) \in \mathcal{A} \quad (33)$$

$$x_{u,v}^i \in \{0, 1\}, \forall i \in V, \forall (u, v) \in \mathcal{A} \quad (34)$$

$$y_{u,v}^{ij} \in \{0, 1\}, \forall i \in V, \forall j \in R, \forall (u, v) \in \mathcal{A} \quad (35)$$

$$\check{a}_{jh} \in \{0, 1\}, \forall j \in R, h = 1, 2, 3, \dots, |\mathbb{O}_j| \quad (36)$$

$$\hat{a}_{jh} \in \{0, 1\}, \forall j \in R, h = 1, 2, 3, \dots, |\mathbb{D}_j| \quad (37)$$

To ensure that the decision variables accurately represent the robotaxi dispatch, several constraints are defined:

- Route Validity:** Constraint (16) ensures that each node is visited at most once, and each arc is passed by at most one robotaxi. Constraint (17) guarantees that s_i is a start node in a route of the robotaxi.
- Robotaxi-request Matching:** Constraints (18)-(21) ensure that passengers in request j are served only once.
- Flow Conservation:** In constraints (20)-(23), for each request, the route is connected, i.e., any intermediate node visited by a robotaxi is preceded by exactly one node and followed by one node. Furthermore, constraints (24) and (25) ensure that a robotaxi carries passengers in request j only between $\bar{o}_j$ and $\bar{d}_j$ (Si et al., 2024).
- PUDO Points:** Constraints (26) and (27) guarantee that each request is assigned exactly one PU point and one DO point.
- Walking Distance:** Constraints (28) and (29) guarantee that the walking distance from an initial point (an origin or a destination) to a recommended PUDO point does not exceed the pre-specified threshold w_{max} .
- BT Offset:** Constraint (30) establishes the lower and upper bounds of u_j , denoted by u^- and u^+ , respectively.
- Detour rate limitation:** Constraint (31) restricts the detour rate of each request below the predefined maximum Δ .
- Timing Constraint:** Constraint (32) ensures that the arrival time τ_i of robotaxi i at $\bar{o}_j$ cannot be later than the new BT t_j , where t_0 is the time that the robotaxi arrives at the last node before $\bar{o}_j$, and $\bar{v}$ denotes the speed of robotaxis;
- Capacity Constraint:** Constraint (33) specifies that the actual number of passengers in a robotaxi must not exceed its seat capacity

q_i .

Additionally, Constraints (34)–(37) give the range of decision variables $x_{u,v}^i$, $y_{u,v}^{ij}$, $\check{a}_{jh}$, and $\hat{a}_{jh}$, respectively.

The aforementioned formulas form the basis of the general model, providing the framework for the robotaxi dispatch. For each stage, specific modifications to the variables and formulas are applied in the corresponding model, as illustrated in Fig. 3. The distinctions between Model 1 and Model 2, relative to the general model, are introduced below.

1) Model 1: It is designed to formulate the first stage. Its decision variables and objective functions align with those of the general model. However, it incorporates specific formulas, as outlined below:

In the first stage, robotaxi dispatch does not account for passenger willingness, implying that all passengers accept the recommended PUDO points and BT. Consequently, $\psi_j=1$ and $\varphi_j=1$, $j \in R$, (7), (9), and (32) can be simplified as follows:

$$\hat{l}_j = \sum_{h=1}^{|\mathcal{O}_j|} \check{a}_{jh} \ell(o_j, o_j^h) + \sum_{h=1}^{|\mathcal{D}_j|} \hat{a}_{jh} \ell(d_j^h, d_j) \quad (38)$$

This simplification further reduces to $\hat{l}_j = \ell(o_j, \bar{o}_j) + \ell(\bar{d}_j, d_j)$.

$$\varepsilon_j = \frac{\lambda |u_j|}{u^+ - u^-} \quad (39)$$

$$t_0 + \frac{\sum_{i=1}^m \sum_{v \in \mathcal{V}} x_{v, \bar{o}_j}^i l(v, \bar{o}_j)}{\bar{v}} \leq t_j + u_j, \forall j \in R \quad (40)$$

2) Model 2: It is designed to formulate the second stage. Its decision variables are $x_{u,v}^i$ and $y_{u,v}^{ij}$. As illustrated in Fig. 3, the values of $\check{a}_{jh}$, $\hat{a}_{jh}$, and u_j , determined from Model 1, are treated as constants in Model 2, where $\check{a}_{jh}^* = \check{a}_{jh}$, $\hat{a}_{jh}^* = \hat{a}_{jh}$, and $u_j^* = u_j$. As the two models are proposed to formulate two stages of the robotaxi problem, both pursue the same goal, i.e., the objective functions of Model 2 are the same as those in Model 1. Its constraints are defined by (16)–(25), (31), (33)–(35), and (45). Specifically, due to $\check{a}_{jh}^* = \check{a}_{jh}$, $\hat{a}_{jh}^* = \hat{a}_{jh}$, and $u_j^* = u_j$, (7)–(10) and (32) are changed to (41)–(45), respectively.

$$\hat{l}_j^* = \psi_j \left(\sum_{h=1}^{|\mathcal{O}_j|} \check{a}_{jh}^* \ell(o_j, o_j^h) + \sum_{h=1}^{|\mathcal{D}_j|} \hat{a}_{jh}^* \ell(d_j^h, d_j) \right) \quad (41)$$

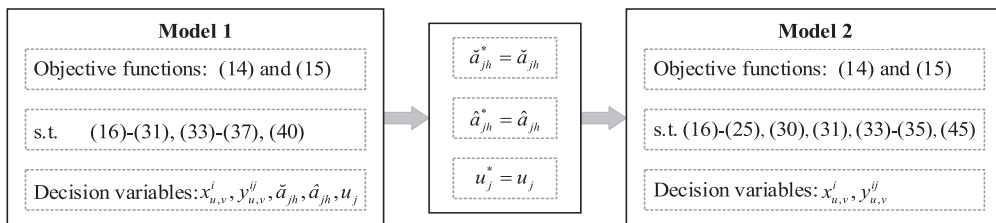
$$\varpi_j^* = \beta \hat{l}_j^* \quad (42)$$

$$\varepsilon_j^* = \frac{\lambda \varphi_j |u_j^*|}{u^+ - u^-} \quad (43)$$

$$e_j = (1 - \gamma_j - \varpi_j^* - \varepsilon_j^*) \hat{e}_j \quad (44)$$

$$t_0 + \frac{\sum_{i=1}^m \sum_{v \in \mathcal{V}} x_{v, \bar{o}_j}^i l(v, \bar{o}_j)}{\bar{v}} \leq t_j + \varphi_j u_j^*, \forall j \in R \quad (45)$$

First, the optimal values of PUDO points and BT offset ($\check{a}_{jh}^*$, $\hat{a}_{jh}^*$, and u_j^*) are directly used in Model 2. Next, the detour rate and its discount are calculated via (5) and (6) according to decision variables $x_{u,v}^i$ and $y_{u,v}^{ij}$. Then, the total detour, including walking and BT offset, is acquired via (41)–(43) according to passenger willingness (ψ_j and φ_j) and those values of $\check{a}_{jh}^*$, $\hat{a}_{jh}^*$, and u_j^* . Ultimately, the travel expense of passengers and the profit per kilometer are computed via (10)–(13), thereby minimizing f_1 and maximizing f_2 through (14)



•Note: “*” indicates that decision variables in Model 1 become constants in Model 2.

Fig. 3. Two models for the two optimization stages, respectively.

and (15).

Existing literature primarily sets the optimization objective to minimize the total cost of customers or maximize the total profit of vehicles (Wang et al. 2024; Si et al., 2024), both of which formulate linear models. While some researchers design non-linear models, they solve those models by developing new algorithms. For example, Ao et al. (2024) propose an approximate Nash equilibrium algorithm to address the non-linear model and prove its non-convexity. Few studies consider the profit per kilometer of vehicles. Since passengers may carpool in our study, compared with the total profit, the profit per kilometer involves both travel revenue and distance, which is a more suitable objective for optimizing robotaxi dispatch.

4. Proposed algorithms

To solve Models 1 and 2 of the RD-RPT, two algorithms are respectively introduced: Nondominated Sorting Genetic Algorithm II via Mass Center (NSGA-MC) and Nondominated Sorting Genetic Algorithm II with Improved Evolution (NSGA-IE). NSGA-MC is developed to solve Model 1 by determining the recommended PUDO points and BT offset through the optimization of $\hat{a}_{jh}$, $\hat{a}_{jh}$, and u_j . A trip similarity determination strategy and specialized evolutionary operations are integrated into the algorithm to enhance the convergence speed. Model 2, in turn, focuses on optimizing routes of robotaxis based on the PUDO points and BT offset determined by Model 1. NSGA-IE is proposed to solve Model 2 by generating optimal values of $x_{u,v}^i$ and $y_{u,v}^j$. This algorithm leverages the outputs of NSGA-MC and incorporates passenger willingness to perform route planning for robotaxis effectively. The following section provides a detailed introduction to the algorithms, where some notations are defined in Table 4.

4.1. Encoding and decoding

Since the robotaxi dispatch problem is a combinatorial optimization problem with multiple constraints, metaheuristic algorithms are employed. Specifically, the encoding process transforms a feasible solution into an individual representation that is suitable for applying genetic operations. The decoding process then maps the individual back into the original solution space, enabling the evaluation of the corresponding objective functions. This design allows the algorithm to efficiently explore the solution space.

4.1.1. Encoding

A solution is a mixed integer string denoted by a triple $\pi=(\pi_1, \pi_2, \pi_3)$, where $\pi_1=(u_1, \dots, u_j, \dots, u_n)$ is a string of real-valued numbers that denotes a sequence of BT offset, and $u_j \in \mathbb{R}$ denotes a BT offset of request j ; $\pi_2=(x_1, \dots, x_j, \dots, x_n, y_1, \dots, y_j, \dots, y_n)$ is an integer string that denotes a sequence of PUDO points, where x_j denotes the index of a recommended PU point in $O_j \cup \{o_j\}$, and y_j denotes the index of a recommended DO point in $D_j \cup \{d_j\}$; and $\pi_3=(\lambda_1, \dots, \lambda_i, \dots, \lambda_m)$ is an integer string that denotes a sequence of routes of robotaxis, where λ_i represents a route without initial location of robotaxi i . For example, when $n=3$, $m=2$, $q_i=4$, and $\eta_j=2$, $i \in \{1, 2\}$, $j \in \{1, 2, 3\}$, a solution is illustrated in Fig. 4. Note that q_i and η_j represent the seat capacity of robotaxi i and the number of passengers in request j , respectively.

4.1.2. Decoding

A decoding process converts a solution into a dispatch plan and computes the values of objective functions. After decoding, a dispatch plan is obtained that includes recommended PUDO points, BT of each request, and routes of robotaxis. The computational

Table 4
Notations in the algorithm.

Notation	Definition
n_p	Population size.
x_j	An index of a recommended PU point in $O_j \cup \{o_j\}$.
y_j	An index of a recommended DO point in $D_j \cup \{d_j\}$.
π_1	A string of real-valued numbers that denotes a sequence of BT offsets.
π_2	An integer string that denotes a sequence of PUDO points.
π_3	An integer string that denotes a sequence of routes of robotaxis.
$\pi=(\pi_1, \pi_2, \pi_3)$	A solution that is a mixed integer string.
Ω_1	Sub-population of BT offset.
Ω_2	Sub-population of the PUDO point.
Ω_3	Sub-population of routes.
$\Omega=(\Omega_1, \Omega_2, \Omega_3)$	A population.
v^k	A velocity of π_1^k , i.e., the direction and length of movement of π_1^k , where k denotes the index of a solution in Ω_1 .
c_m	A mass center of an initial population.
θ_1	Self-learning rate, $\theta_1 \in (0, 1)$.
θ_2	Global learning rate, $\theta_2 \in (0, 1)$.
λ_i	A route without the initial location of robotaxi i .
G	The maximum iteration count.
p_1	A probability to update π_1 .
p_2	A probability to update π_2 .
p_3	A probability to update π_3 .

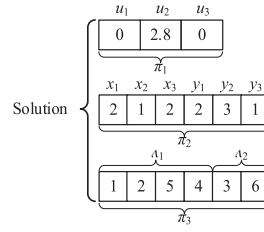


Fig. 4. An example of encoding.

steps are summarized as follows:

Step 1: Divide a solution into three fragments: π_1 , π_2 , and π_3 ;

Step 2: Update BT of each request via π_1 , $t_j = t_j + u_j$, and get a discount for BT;

Step 3: Update PUDO points of each request via π_2 and get a discount for passengers for walking;

Step 4: Divide π_3 into m routes, assign robotaxis to requests via π_3 , and let robotaxis follow a planned route λ_i to visit a specified point in each set of PUDO points; and

Step 5: Compute a detour rate and the corresponding discount, and calculate travel expense e_j for request j and the total travel cost c for robotaxis.

NSGA-MC pre-generates routes of robotaxis to estimate the optimality of the recommended PUDO points and BT. Consequently, the solution π in the algorithm consists of fragments π_1 , π_2 , and π_3 . In NSGA-IE, the PUDO points and BT are fixed with passenger willingness incorporated. Here, the solution obtained by NSGA-IE is denoted by π_3 . As the definition of π_3 is the same as that in NSGA-MC, we will not present the details of the encoding and decoding in NSGA-IE.

4.2. NSGA-MC

Fig. 5 depicts the framework of NSGA-MC. First, parameters, such as n_p , G , θ_1 , θ_2 , p_1 , p_2 , and p_3 , are set. Then, the population is initialized, where Part 1 provides a trip similarity determination strategy, which is proposed to assess the probability of carpooling among requests. The maximum allowable distance between two origins/destinations is denoted by d_l , and the maximum allowable time interval is represented by τ . Two trips of request j and j' are defined as similar, i.e., $M_{jj'} = 1$, if the following conditions are

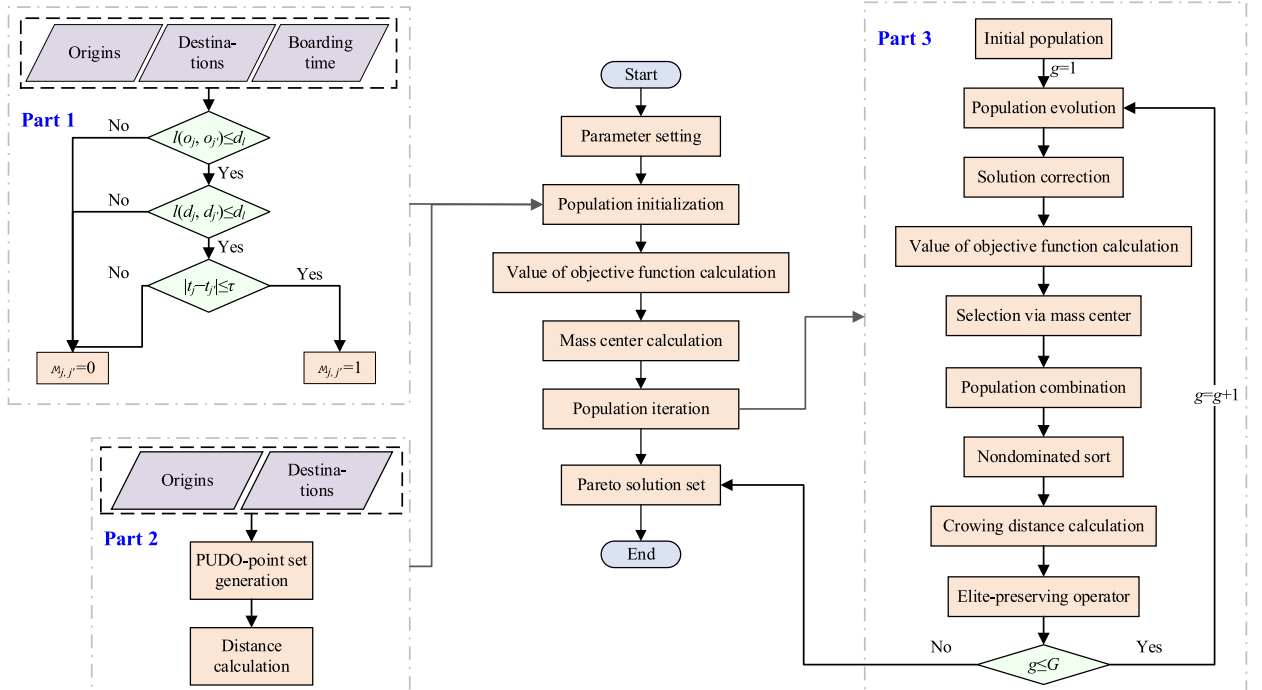


Fig. 5. Flowchart of NSGA-MC.

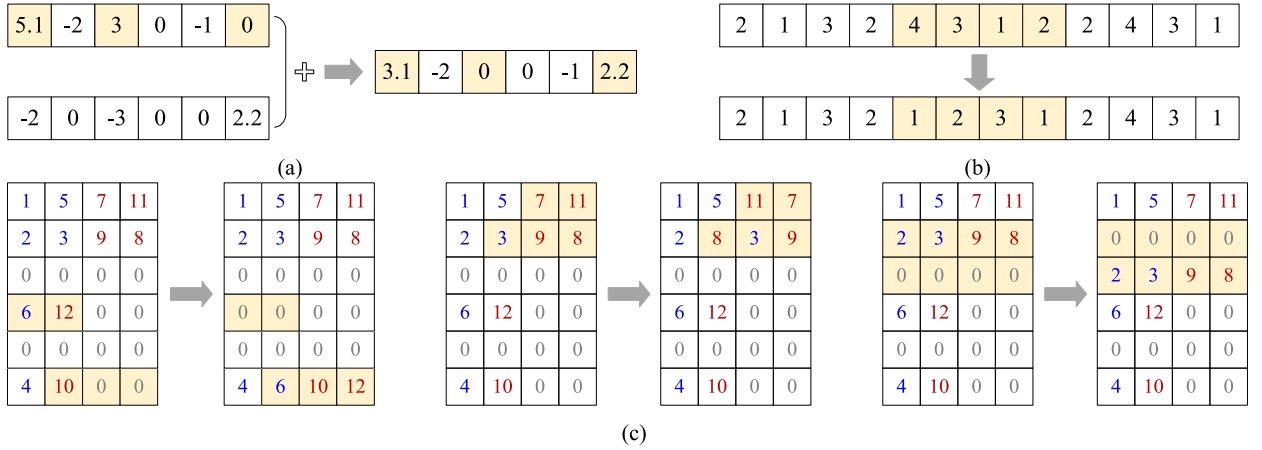


Fig. 6. Cases of evolution: (a) π_1 -evolution; (b) π_2 -evolution; and (c) π_3 -evolution.

satisfied: 1) $l(o_j, o_f) \leq d_l$ (the distance between their origins is no more than d_l); 2) $l(d_j, d_f) \leq d_l$ (the distance between their destinations is no more than d_l); and 3) $|t_j - t_f| \leq \tau$ (the time interval between their boarding time is no more than τ). Otherwise, $\mathcal{N}_{j\bar{j}} = 0$. Additionally, the sets of PUDO points are generated according to the origins and destinations of requests, while the distance between any two points is computed in Part 2. Upon completing the aforementioned steps, the algorithm initializes a population, thereby computing the values of objective functions for each solution and the mass center of the initial population. The details are shown in Section 4.2.1. Next, the population evolves iteratively until a termination criterion is met, i.e., Part 3, where the strategies for updating feasible solutions are defined in Section 4.2.2. During the iteration, the feasible solutions are selected via the mass center, as depicted in Section 4.2.3. Finally, a Pareto-optimal set is formed.

4.2.1. Population initialization

Algorithm 1 describes the initialization of solutions, referred to as individuals. This is a key step in generating a feasible solution. A solution $\pi = (\pi_1, \pi_2, \pi_3)$ consists of three fragments. The three sub-populations of BT offset, PUDO point, and routes of robotaxis are defined as Ω_1 , Ω_2 , and Ω_3 , respectively. A population is denoted by $\Omega = (\Omega_1, \Omega_2, \Omega_3)$.

First, the initial Ω_1 and its velocity v^k are obtained via Lines 2–4. Then, Ω_2 is initialized, i.e., the PUDO points are randomly selected in their sets of PUDO points via Lines 5–19. Next, Ω_3 is initialized, i.e., the driving routes of robotaxis satisfying the constraints are randomly initialized via Lines 20–26. Finally, the three sub-populations are merged to form Ω , and a mass center c_m is calculated via Line 29. Note that the mass center is denoted by:

$$c_m = \left(\sum_{k=1}^{n_p} \frac{f_1^k}{n_p}, \sum_{k=1}^{n_p} \frac{f_2^k}{n_p} \right) \quad (46)$$

where f_1^k and f_2^k represent the values of two objective functions of π^k in the initial population, respectively.

4.2.2. Three-operation evolution

This section presents three operations for population evolution. Each operation is designed to update one sub-population. Continuous coding is adopted in Ω_1 . PSO is suitable for solving continuous optimization problems (Bonyadi and Michalewicz, 2016). Therefore, a velocity v_{g+1}^k is used to update fragments in a solution, which is designed based on a velocity update operation in PSO. The velocity is defined as:

$$v_{g+1}^k = \theta_1 v_g^k + \theta_2 \cdot \text{rand} \cdot (v_g^k - v_g^*) \quad (47)$$

where v_g^k denotes a velocity of π_1^k in the g -th iteration, v_g^* denotes the velocity of a Pareto solution, $\text{rand} \in [0, 1]$ is a random real number, and $\theta_1 \in (0, 1)$ and $\theta_2 \in (0, 1)$ denote the self-learning rate and global learning rate, respectively.

In Ω_1 , new fragments are generated by $\pi_1^k + v^k$. In addition, an integer coding is adopted in Ω_2 and Ω_3 . Ω_2 generates new fragments with probability p_2 after generating a new Ω_1 . Ω_3 generates new fragments with probability p_3 after generating a new Ω_2 . The details are presented in Algorithm 2. Lines 3–11 show the updating process of π_1^k . An example is shown in Fig. 6 (a), where the first and second lines to the left of the right parenthesis represent π_1^k and v^k , respectively, while the right side shows the evolved π_1^k . The yellow

background indicates that the values are changed. Lines 12–29 illustrate the evolution of π_2^k , as shown in Fig. 6 (b). Lines 30–43 explain the evolution of π_3^k . Fig. 6 (c) shows three examples. Note that π_3 is represented as a matrix to make its evolution clear. Each row represents a route of the robotaxi. Unlike the third fragment in Fig. 4, the ones presented here are fully encoded, with '0' indicating no requests. The blue and red fonts represent PUDO points, respectively. The third fragment in Fig. 4 shows a simplified version of π_3 .

Algorithm 1 Population Initialization

Input: A set of robotaxis V , a set of requests R , population size n_p , and the sets of PUDO points $\mathcal{O}_j$ and $\mathcal{D}_j, j \in \mathbf{N}_n^+$.

Output: Initial population Ω .

```

1   for each  $k \in [1, n_p]$  do
2       Generate  $\pi_1^k$  randomly satisfying (30); //initialize fragment in sub-population  $\Omega_1$ 
3       Generate an initial velocity  $v^k$  of  $\pi_1^k$ ;
4        $\pi_1^k = \pi_1^k + v^k$ ;
5       for each  $j \in [1, n]$  do //initialize fragment in sub-population  $\Omega_2$ 
6           Generate random number  $\lambda_1, \lambda_1 \in (0, 1)$ ;
7           if  $\lambda_1 < p$  then //  $p$  is a probability that a PUDO point is changed
8                $\pi_2^k(j) = 1$ ;
9           else
10               $\pi_2^k(j)$  is an integer,  $\pi_2^k(j) \in [2, |\mathcal{O}_j|]$ ;
11          end
12          Generate random number  $\lambda_2, \lambda_2 \in (0, 1)$ ;
13          if  $\lambda_2 < p$  then
14               $\pi_2^k(j+n) = 1$ ;
15          else
16               $\pi_2^k(j+n)$  is an integer,  $\pi_2^k(j+n) \in [2, |\mathcal{D}_j|]$ ;
17          end
18      end
19      Obtain  $\pi_2^k$ ;
20      Divide  $\mathbf{N}_n^+$  into  $m$  sequences; // the sequence is allowed to be empty
21      for each  $i \in [1, m]$  and the  $i$ -th sequence is not empty do
22          //initialize fragment in sub-population  $\Omega_3$ 
23          Sort the  $i$ -th sequence according to BT;
24          Add the corresponding DO points  $\mathbf{N}_{2n}^+ \setminus \mathbf{N}_n^+$  in the  $i$ -th sequence via (24)-(25);
25          Move the elements that violate the constraints (31)-(33) in the  $i$ -th sequence;
26      end
27      Obtain  $\pi_3^k$ ;
28       $\pi^k = (\pi_1^k, \pi_2^k, \pi_3^k)$ ;
29  end
30  Obtain  $\Omega = (\Omega_1, \Omega_2, \Omega_3)$  and calculate its mass center  $c_m$ ;
31  Return.

```

4.2.3. Intra-iteration selection

Algorithm 2 Three-operation Evolution**Input:** Ω, p_1, p_2 , and p_3 **Output:** Ω' // a new population

```

1  for each  $k \in [1, n_p]$  do
2       $z=0$ ;
3      if  $rand < p_1$  then // updating  $\pi_1$ 
4          Calculate  $v^k$  via (47);
5           $\pi_1^k = \pi_1^k + v^k$ ;
6          if  $\pi_1^k < u^-$  or  $\pi_1^k > u^+$  then
7               $\pi_1^k = u^- + rand(u^+ - u^-)$ ;
8          end
9           $z=1$ ;
10     end
11     Obtain a new  $\pi_1^k$ ;
12     if  $rand < p_2$  then // updating  $\pi_2$ 
13         if  $z=0$  then
14             Generate an integer  $n'$ ,  $0 < n' \leq 2n$ ;
15             Select a sequence  $\pi_2'$  of length  $n'$  from  $\pi_2^k$ ;
16             for each  $j' \in [1, n']$  do
17                 if  $rand < p_2$  then
18                     if  $j' \leq n$  then
19                          $\pi_2'(j') = x_1$ ,  $1 < x_1 \leq |O_j|$  and satisfying (28); //  $|O_j|$  represents the cardinality of  $O_j$ 
20                     else
21                          $\pi_2'(j') = x_2$ ,  $1 < x_2 \leq |D_{(j'-n)}|$  and satisfying (29); //  $|D_{(j'-n)}|$  represents the cardinality of  $D_{j'-n}$ 
22                     end
23                 else
24                      $\pi_2'(j') = 1$ ;
25                 end
26             end
27         end
28     end
29     Obtain a new  $\pi_2^k$ ;
30     Divide  $\pi_3^k$  into  $m$  routes denoted by  $A=[A_i]_{m \times 1}$ ;
31     Correct  $A_i$  in  $A$  if (31) or (32) is violated by  $\pi_1^k$  and  $\pi_2^k$ ;
32     if  $rand < p_3$  then // updating  $\pi_3$ 
33         Reorder  $A_i$  in  $A$ ;
34         Generate an integer  $m'$ ,  $m' < m$ ;
35         for each  $i \in [1, m']$  do
36             Remove elements with a value of 0 in  $A_i$ ;
37             if  $|A_i| > 2$  then
38                 Swap two nodes in  $A_i$ ;
39                 Correct  $A_i$  if (24), (25), (31), (31), or (33) is violated; // if a constraint is violated and it cannot be satisfied by
                    adjusting one route, the nodes are moved from one route to another that satisfies the constraint
40             end
41         end
42     end
43     Obtain a new  $\pi_3^k$ ;
44      $\pi^k = (\pi_1^k, \pi_2^k, \pi_3^k)$ ;
45 end
46 Obtain new  $\Omega'_1, \Omega'_2$ , and  $\Omega'_3$ ;
47  $\Omega' = (\Omega'_1, \Omega'_2, \Omega'_3)$ ; //  $\Omega'$  denotes a new population
48 Return.

```

An NSGA-II selects solutions to obtain a new population through nondominated sort, crowding distance calculation, and elite-preserving operation. However, the values of the Pareto front obtained based on NSGA-II are scattered. Some Pareto solutions perform well on one objective but poorly on others. RD-RPT involves two objectives. One objective is to minimize the travel expense of

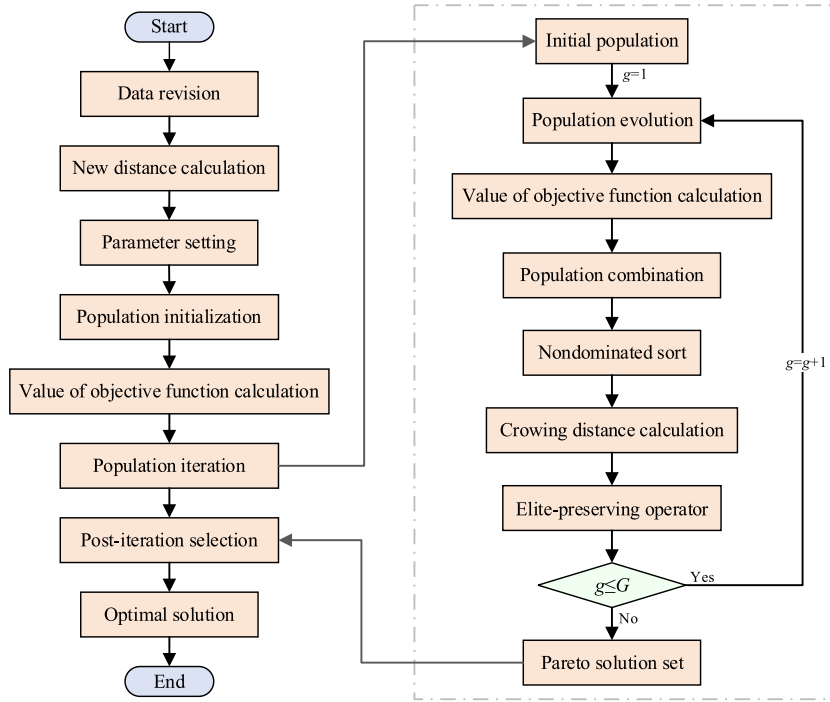


Fig. 7. Flowchart of NSGA-IE.

passengers. Poor performance in this objective may lead to decreased passenger satisfaction and a lower likelihood of choosing robotaxis in the future. The other objective is to maximize the profit per kilometer of robotaxis. Poor performance in this objective may reduce the profit of the robotaxi company, potentially leading to operational difficulties. Therefore, a comprehensive consideration of both objectives is necessary, aiming to select solutions that outperform the initial values on both objectives.

The mass center of the initial population is obtained using a strategy similar to determining a cluster center in clustering. The values of the mass center on the two objectives are used as a criterion for solution selection. Such operations as nondominated sort, crowding distance calculation, and elite-preserving operation are used to select individuals, thereby generating a new population during iteration. To choose a high-quality solution, a selection strategy is proposed and detailed as follows:

- 1) Calculate the value of the objective function for each individual;
- 2) Delete the individuals whose f_1 is larger than the mass center, or f_2 is less than the mass center in the Pareto set; and
- 3) Reserve others for next operations, as shown in Fig. 5.

4.3. NSGA-IE

NSGA-IE is a multi-objective optimization algorithm that is proposed to solve Model 2. First, its inputs are new PUDO points and BT obtained by NSGA-MC, and the distances among points are calculated. Next, the parameters of the algorithm are set. Then, the population is initialized. After that, the values of the objective functions for the initial population are calculated. Finally, the algorithm enters the iterative process to obtain the optimal route. The details of NSGA-IE are illustrated in Fig. 7. In the flowchart, the evolution operator is a key step, which is detailed in Algorithm 3. Lines 2–10 show the process of internal conversion. Specifically, the positions of PUDO points in some routes are swapped. Lines 11–16 are designed to promote passenger carpooling, which merges some routes of robotaxis with other feasible ones. Ultimately, Lines 17–20 illustrate an external exchange by swapping two routes between two robotaxis, thereby making the initial location of the robotaxi close to its next node in the route.

After the termination of the iteration, a Pareto-optimal set is obtained. However, in multi-objective optimization problems, only one feasible solution is typically adopted for implementation in the real world. Therefore, it is necessary to select an optimum from the Pareto-optimal set as the final one. To balance the two conflicting objectives, a post-iteration selection is proposed. The details are below:

Algorithm 3 Evolution operator

Input: An initial population Ω_3 and updating probability p_r .

Output: New population Ω'_3 .

```

1  for each  $\pi_3 \in \Omega_3$  do
2      if  $\text{rand} < p_r$  then //internal conversion
3          Select a segment  $\pi_{\text{seg}}$  of  $\pi_3$  including  $m'$  non-empty routes of robotaxis;
4          Divide  $\pi_{\text{seg}}$  into  $m'$  routes denoted by  $A=[A_i]_{m' \times 1}$ ;
5          for each  $i \in [1, m']$  do
6              Swap  $\bar{o}_j$  and  $\bar{d}_{j'}$  in  $A_i$ ;
7              if  $A_i$  violates constraints (24)-(25) and (31)-(33) then
8                  Correct  $A_i$  by adjusting the positions of the points;
9              end
10         end
11     else if  $p_r \leq \text{rand}$  and  $\text{rand} \leq p_r + (1-p_r)/2$  then //request recombining
12         for each  $i \in [1, m]$  do
13             if robotaxi  $i$  serves only one request then
14                 Merge its route with other feasible ones in  $\pi_3$ ;
15             end
16         end
17     else //external exchange
18         Divide  $\pi_3$  into  $m$  routes denoted by  $A=[A_i]_{m \times 1}$ ;
19         Swap  $A_i$  and  $A_{i'}$  in  $\pi_3$ ;
20     end
21 end
22  $\Omega'_3 = \Omega_3$ ;
23 Return.

```



Fig. 8. Manhattan map.

Post-iteration selection: 1) The medians of the objective functions f_1 and f_2 are computed, respectively, which together form a virtual intermediate solution; 2) The distance between each solution on the Pareto front and the virtual intermediate one is calculated; and 3) The solution closest to the virtual intermediate one is selected as the final solution.

Given that PUDO points and BT are fixed in Model 2, which are variables in Model 1, the search space of NSGA-IE becomes smaller than that of NSGA-MC. Consequently, NSGA-IE no longer selects solutions via a mass center. It also refines NSGA-II by proposing an evolution operator similar to the update strategy for Ω_3 used in NSGA-MC to generate new solutions. In summary, the proposed algorithms attain PUDO points and BT offset via NSGA-MC and generate optimal routes of robotaxis via NSGA-IE.

5. Experimental results

In this section, experiments based on real datasets are presented to verify the validity of the proposed model and the performance of our algorithms. Firstly, the experimental environment, datasets, and parameter settings are presented. A small-scale instance is conducted to validate the proposed model. Then, the trip similarity among requests depicts the potential for passengers to engage in carpooling. Next, the proposed algorithm is compared with NSGA-II and MOEA/D, and the performance of our method is demonstrated through ablation. Finally, an analysis is conducted on the impact of passenger willingness on the performance of the proposed method.

5.1. Experimental setup

A taxi trip record dataset in Manhattan, New York (New York City Taxi and Limousine Commission, 2010) is employed along with the imported road network of Manhattan, as shown in Fig. 8. Data preprocessing is implemented in PyCharm 2022.1.2 and sumo-1.14.1. All algorithms are implemented in MATLAB R2022a, and CPLEX 12.10 is used as the commercial solver to validate the proposed mathematical model. The operating environment is Windows 11 with 12th Gen Intel (R) Core (TM) i7-12700H CPU (2.30 GHz/16.00 GB RAM). In CPLEX, the time limit is set to 120 s to ensure that feasible solutions can be obtained; the MIP gap tolerance is set to 0.01%, which corresponds to a widely used solver configuration; and up to six threads are allowed.

Firstly, the map of Manhattan is imported into Simulation of Urban Mobility (SUMO) (Lopez, et al., 2018), a traffic simulation tool, and then a Traffic Control Interface (TraCI) package is imported into PyCharm to generate sets of PUDO points:

- 1) An edge close to an initial origin/destination is identified, and its incoming and outgoing edges are discovered;
- 2) Points on these edges that are 30 m away from the intersection and less than 300 m from the requests' origin/destination are considered as new PUDO points; and
- 3) A set of PUDO points is formed by the initial origin or destination and the new PUDO points.

The driving distance between PUDO points in the road network is calculated. For simplicity, Euclidean distance is adopted as a walking distance between two points in the same set of PUDO points. The distance between two points that do not belong to the same set of PUDO points is the driving distance of robotaxis. Note that the robotaxi needs to drive in the given direction to obey the traffic rules. Thus, the driving distance is calculated according to the real road network. For most streets in Manhattan, the legal speed limit is 25mph (40 km/h) (Wazeopedia, 2024). In several pedestrian-dense areas, the speed limit is about 30 km/h. Therefore, an average speed of 30 km/h is set for robotaxis over safety concerns in the following experiments. The number of requests equals the maximum number of available robotaxis. At most, two requests will be allowed to carpool together. The values of key parameters are presented in

Table 5
Parameter setting.

Parameters	Values	Parameters	Values
n_p	200	e_0	3
G	2000	c_r	2.18
p_1	0.3	c^-	0.4
p_2	0.4	α	0.1
p_3	0.8	β	0.2
u^-	-10	λ	0.2
u^+	10	γ_0	0
w_{max}	0.3	θ_1	0.8
Δ	2	θ_2	0.5

Table 6
Four request densities.

No.	Time interval	The number of requests
R_1	1 min	39
R_2	10 min	80
R_3	30 min	100
R_4	60 min	100

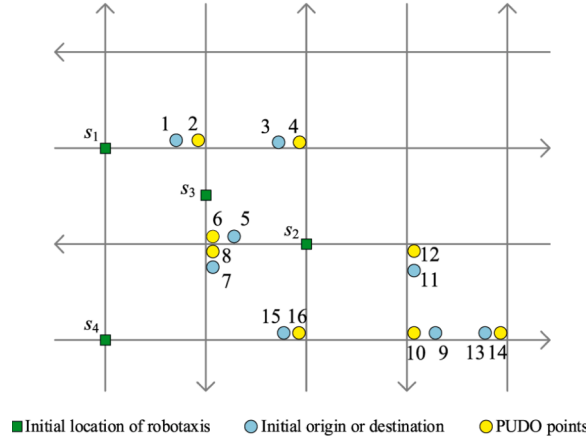


Fig. 9. Locations of robotaxis and passengers.

Table 7
The information on requests.

Request	BT	Origin	Destination	PU point	DO point
r_1	9: 00 am	1	9	2	10
r_2	9: 00 am	3	11	4	12
r_3	9: 10 am	5	13	6	14
r_4	9: 10 am	7	15	8	16

Table 8
Comparison of the exhaustive method, CPLEX, and the proposed algorithm.

Algorithm	$f_1 \downarrow$	$f_2 \uparrow$	$gap_1 \downarrow (\%)$	$gap_2 \uparrow (\%)$	Computation time $\downarrow$ (s)	Amount of used robotaxi $\downarrow$
Exhaustive Method	56.5608	2.5926	1.2896	70.8580	1095.5827	3
CPLEX	55.8407	1.5174	—	—	13.0837	4
Proposed Algorithm	55.8407	2.0279	0	33.6431	15.9636	2

● Note: $gap_k = (f_k - f_{CPLEX}^k) / f_{CPLEX}^k \times 100\%$, $k \in \{1, 2\}$, where gap_k is the optimality gap, differing from the MIP gap tolerance of CPLEX; f_k represents the value of the objective function; and f_{CPLEX}^k represents that obtained by CPLEX.

Table 5. A request density represents the number of requests in different time intervals. Four groups of data under varying request densities in Table 6 are tested to simulate low- and high-demand scenarios at the city level over a fixed time interval.

5.2. Small-scale instance

Exact algorithms cannot efficiently handle large-scale instances within limited time frames. To verify the validity of the proposed

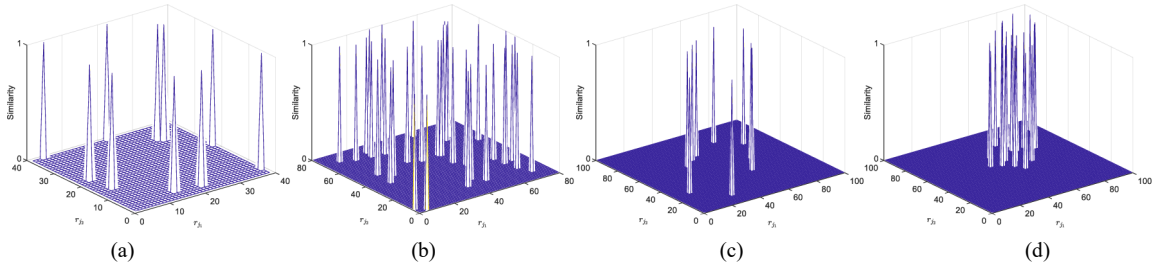


Fig. 10. Trip similarity among requests in four groups of data: (a) trip similarity of R_1 ; (b) trip similarity of R_2 ; (c) trip similarity of R_3 ; and (d) trip similarity of R_4 .

model and the performance of the proposed algorithm, a small-scale instance is employed. The solutions obtained by the proposed algorithm are compared with those derived from an exhaustive search method and CPLEX. The instance involves four robotaxis and four requests, denoted by v_1 - v_4 and r_1 - r_4 , respectively, with their locations illustrated in Fig. 9. s_1 - s_4 represent the initial locations of the robotaxis. Blue dots indicate the initial origins or destinations for passengers, while yellow dots represent candidate PUDO points. For instance, the origin of r_1 is dot 1, with its new PU point designated as dot 2. These points form a set of candidate PU points in r_1 . Additional request details are provided in Table 7.

Table 8 presents the solutions obtained from the exhaustive method, CPLEX, and the proposed algorithm for the small-scale instance. According to the definition of f_2 , the proposed mathematical model is non-convex and non-linear. As a result, it cannot be directly solved using commercial solvers. When employing CPLEX, the objective function is transformed into a linear form, where minimizing the total travel cost of robotaxis is adopted as the optimization objective. The travel expense of passengers (f_1) is converted into a constraint, and the value of f_2 can be indirectly computed accordingly. It can be observed that the exhaustive method and the proposed algorithm yield Pareto solutions, while the solution obtained by CPLEX is dominated by that of the proposed algorithm. The results demonstrate that the proposed algorithm provides an approximately optimal solution within a finite time. Additionally, our algorithm saves more robotaxis than others, which illustrates that the algorithm is conducive to promoting carpooling. This finding confirms the validity of the proposed model and demonstrates that the proposed algorithm effectively determines high-quality solutions for the problem.

5.3. Trip similarity among requests

The values of $S = [\mathcal{M}_{j_1, j_2}]_{n \times n}$ capture the similarity of trips between two requests. The similarity of the four groups of data is depicted in Fig. 10. The x-axis and y-axis represent the two requests, respectively, and the z-axis represents the value of $\mathcal{M}_{j_1, j_2}$. In Fig. 10, compared with (a) and (c), (b) and (d) show that there are more requests with similar trips on R_2 and R_4 , respectively. A greater

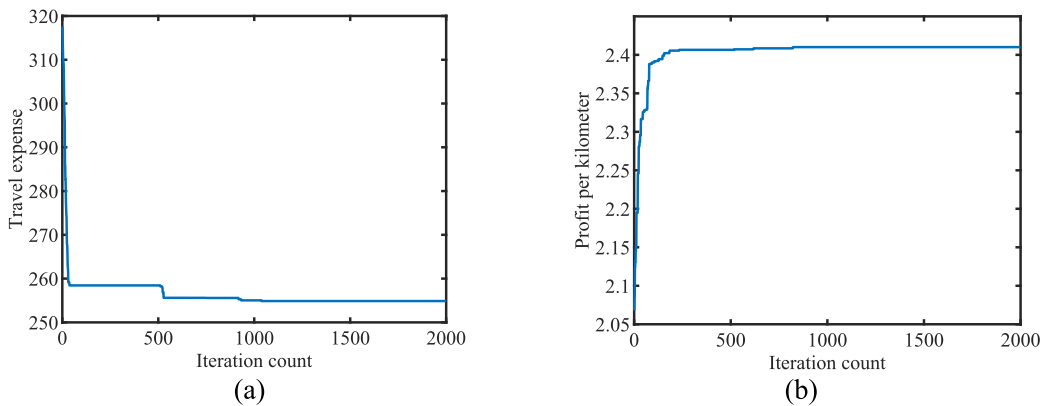


Fig. 11. Iterative process of the proposed algorithm: (a) Travel expense of R_1 ; and (b) Profit per kilometer of R_2 .

number of similarity values equal to 1 indicates more potential passengers for carpooling. Note that S is a symmetric binary matrix, where $\mathcal{M}_{j_1, j_2} = 1$ if two requests are similar and $j_1 \neq j_2$; otherwise, $\mathcal{M}_{j_1, j_2} = 0$. $\mathcal{M}_{j_1, j_2} = 0$ if $j_1 = j_2$, because $j_1 = j_2$ indicates that requests j_1 and j_2 are the same one. The similarity can serve as a potential factor for carpooling in the route planning of robotaxis, thereby generating an enhanced initial solution in the first stage.

5.4. Convergence of the proposed algorithm

The convergence of iterative algorithms is a pivotal metric for evaluating optimization algorithms. Given the vast search space, the convergence analysis of NSGA-MC is necessary. As depicted in Fig. 11 and Fig. A.1 of Appendix A, Fig. 11(a) and Figs. A.1(a), (c), and (e) correspond to the value of f_1 on R_1 - R_4 , respectively; and Fig. 11(b) and Figs. A.1(b), (d), and (f) correspond to the value of f_2 on R_1 - R_4 , respectively. In most cases, the algorithm converges within 1000 iterations. As the number of requests increases, the convergence speed of the algorithm gradually decreases. The value of f_1 correlates positively with the number of requests, whereas f_2 correlates negatively. Due to the varying initial origins, destinations, and BT in requests, the initial values of objective functions for each group of data are also different. The two objectives of this algorithm are to minimize f_1 and maximize f_2 . These curves show that as the iteration count increases, the value of f_1 decreases and the value of f_2 increases, with both tending to converge eventually.

5.5. Algorithm evaluation through performance metrics

There are two algorithms to solve RD-PRT, which indicates the solving process consists of two consecutive optimizations, where NSGA-MC and NSGA-IE are proposed to solve Models 1 and 2, respectively. To ensure that all algorithms are a single optimization, this section analyzes the results obtained by solving Model 1 with NSGA-MC, NSGA-II, and MOEA/D. Two performance metrics are used to illustrate the superiority of the proposed algorithm for solving an RD-PRT. Multi-objective optimization algorithms often get a set of approximate solutions. The following metrics are used to evaluate these solutions (Li and Yao, 2019).

1) Hypervolume (HV) (Shang et al., 2021): It provides the volume of the hypercube formed by the Pareto set and reference points in the objective space. The metric is defined as:

$$HV = \bigcup_{i=1}^{|P^*|} \hat{v}_i \quad (48)$$

where $\hat{v}_i$ is the volume of the hypercube formed by a Pareto set and reference point, and P^* is the optimal solution set. This metric is used to evaluate the convergence of the Pareto set. The larger the metric value, the better the performance of the algorithms.

2) Inverted Generational Distance (IGD) (Sun et al., 2019): It calculates the average distance among the individuals in the optimal solution set and Pareto set. The metric is formulated by:

$$IGD = \frac{\sum_{p \in P^*} \text{dist}(p, PF)}{|P^*|} \quad (49)$$

where PF is a Pareto set generated by an algorithm, and $\text{dist}(p, PF)$ is the nearest distance from p to solutions in PF . This metric can evaluate the approximation degree of the Pareto set and the optimal solution set, and reflect the diversity and convergence of the Pareto set to some extent. The lower the metric value, the better the performance of the algorithm.

Table 9 presents the HV and IGD values obtained by NSGA-MC, NSGA-II, and MOEA/D on R_1 - R_4 . To calculate HV, the reference points are set as [335, 1.80], [730, 1.70], [930, 1.55], and [1000, 1.4] for R_1 - R_4 , respectively. Values in bold indicate a significant improvement over the others. The HV obtained by this algorithm is greater than that of the other algorithms. The proposed algorithm on R_2 - R_4 achieves the lowest IGD, outperforming the others. The results demonstrate that our algorithm exhibits better diversity and convergence than NSGA-II and MOEA/D in solving this problem in most request diversities. Although the IGD value obtained by NSGA-MC in R_1 is higher than that of the other algorithms. In R_1 , due to the high density of requests within a short time window, the solutions in the Pareto set obtained by NSGA-MC have the values of the objective functions close to each other. Therefore, the average of the nearest distances from the solutions in the optimal solution set to those in the Pareto set increases, which is reflected in a higher IGD value. However, the algorithm remains competitive in other instances. Additionally, the variances of the HV obtained from each algorithm are low and are not significantly different. The proposed algorithm surpasses others in terms of the variances of IGD except R_1 , indicating enhanced stability in most instances. These results demonstrate that the proposed algorithm achieves superior performance across the evaluated metrics, showcasing its diversity, convergence, and stability when addressing the problem.

Table 9

Comparison of experimental results on the three algorithms via HV and IGD.

Metric	No.	NSGA-MC		NSGA-II		MOEA/D	
		mean	variance	mean	variance	mean	variance
HV	R_1	0.18231	2.96E-05	0.17961	2.07E-05	0.17956	1.33E-05
	R_2	0.15773	1.06E-05	0.15284	1.25E-05	0.10671	2.62E-05
	R_3	0.17109	1.25E-05	0.16962	7.90E-06	0.13361	1.16E-05
	R_4	0.14578	7.96E-06	0.14472	1.15E-05	0.12229	7.94E-06
IGD	R_1	0.35200	0.12886	0.20802	0.01532	0.28494	0.05793
	R_2	0.50125	0.27341	1.76748	1.62644	32.25906	7.56674
	R_3	0.25957	0.00722	0.28369	0.01771	16.81520	5.28508
	R_4	0.18712	0.00755	0.79193	0.87783	15.51838	4.24356

5.6. Ablation study

In most studies, robotaxi dispatch only considers the robotaxi-request matching and the route planning of robotaxis, i.e., robotaxi dispatch without recommendation (RD-NR). Some studies consider changing the PUDO points of passengers, i.e., the robotaxi dispatch via the recommendation of PUDO points (RD-RP). This work innovatively proposes to change the BT of passengers, i.e., the robotaxi dispatch via the recommendation of BT (RD-RT). An ablation study is conducted on the three aspects mentioned above and compared with the proposed method, i.e., the robotaxi dispatch via the recommendation of PUDO points and BT (RD-RPT).

Fig. 12 demonstrates the experimental results of different models. Overall, as the request density decreases, there is an observable trend toward higher average travel expense of passengers, coupled with a decline in profit per kilometer of robotaxis. The greater the walking distance or time offset, the larger the discount enjoyed by passengers, thereby reducing their travel expense. The average travel expense of RD-RPT model is 1.03, 0.38, and 0.65 less than that of the RD-RN, RD-RT, and RD-RP models, respectively. Correspondingly, the profit per kilometer experiences a slight dip, with the decrement ranging in [0.22, 0.54], averaging at 0.32. In this work, the cost per kilometer of robotaxis is calibrated to 0.4 (i.e., $c^- = 0.4$), while the profits accrued through this method far exceed c^- . These results demonstrate the viability of our method in alleviating the travel expense of passengers, while preserving the profitability of robotaxi service.

According to the mathematical models, the travel expense of passengers is influenced by the travel distance, walking distance, time offset, and detour. Similarly, the profit per kilometer of robotaxis is determined by travel cost, travel revenue (i.e., total travel expense of passengers), and driving distance of robotaxis. A decline in travel expense is identified as a critical factor contributing to the reduction in the profit per kilometer of robotaxis. The results of the ablation study corroborate this observation. The recommendation of PUDO points or BT introduces a relative discount, thereby lowering the travel expense of passengers and subsequently reducing the profit per kilometer of robotaxis compared to RD-NR. From the perspective of pilot deployment and the long-term development of robotaxis, this outcome may be considered acceptable.

5.7. Sensitivity analysis

The passenger willingness to accept the recommendation of BT is denoted by φ_j , while the willingness to accept the recommendation of PUDO points is denoted by ψ_j . To simulate φ_j and ψ_j taking values 1 or 0, this section examines various recommendation acceptance rates of BT or PUDO points for requests in R_1 - R_4 , calculated by $\frac{1}{n} \sum_{j=1}^n \varphi_j$ and $\frac{1}{n} \sum_{j=1}^n \psi_j$, respectively. The study encompasses the full spectrum of acceptance rates, from 0% to 100%, with analysis conducted at intervals of 0%, 25%, 50%, 75%, and 100%. To minimize the influence of randomness, the recommendation acceptance rate of BT is set equal to that of PUDO points. In Fig. 13 and Fig. A.2, the x-axis represents the recommendation acceptance rate. For brevity, Fig. A.2 is provided in Appendix A. Fig. 13(a) and Figs. A.2(a), (d), and (g) illustrate the impact of varying acceptance rates on the total travel expense. As the acceptance rate increases, the total travel expense decreases as the proportion increases. The results in Fig. 13(b) and Figs. A.2(b), (e), and (h) demonstrate that as the recommendation acceptance rate increases, the profit per kilometer of robotaxis exhibits a declining trend, though it remains significantly higher than the cost per kilometer of robotaxis. Furthermore, the study analyzes the percentage of requests with a negative detour rate, defined as the ratio of requests where the travel distance from the actual PU point to the DO point is less than the original distance between the origin and destination, relative to the total number of requests, denoted by:

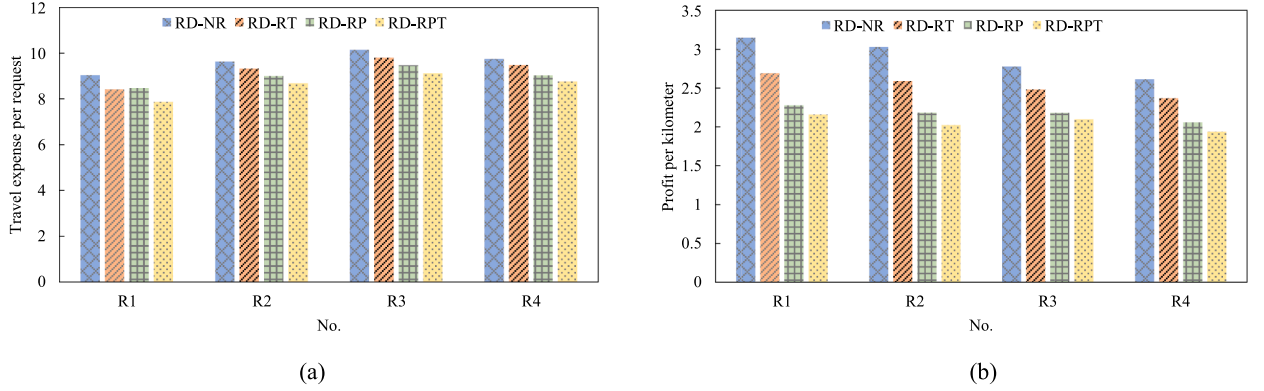


Fig. 12. Results of ablation study: (a) Travel expense and (b) Profit per kilometer.

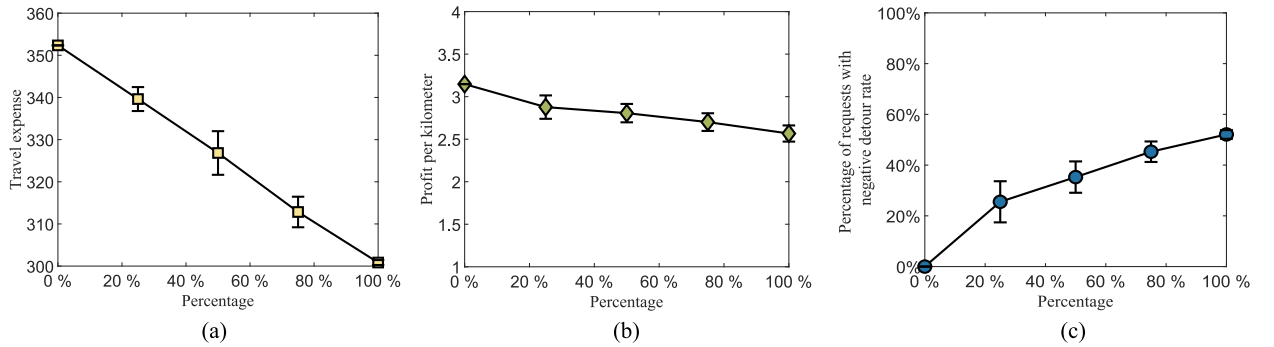


Fig. 13. The impact of varying acceptance rates on recommendation: (a) Travel expense of R_1 ; (b) Profit per kilometer of R_1 ; and (c) Percentage of requests with a negative detour rate of R_1 .

$$p_{nd} = \frac{1}{n} \sum_{j=1}^n \mathbb{I}(\delta_j \leq 0) \times 100\% \quad (50)$$

where $\mathbb{I}(\bullet)$ is an indicator function that returns 1 if the condition is true and 0 otherwise. δ_j represents the detour rate of request j . Fig. 13(c) and Figs. A.2(c), (f), and (l) reveal an increase in the number of requests exhibiting a negative detour rate as the recommendation acceptance rate rises, illustrating a positive correlation between the acceptance rate and the percentage of requests with a negative detour rate.

By varying the values of φ_j and ψ_j , the impact of passenger willingness on the robotaxi dispatch is analyzed. The results demonstrate that with a larger recommendation acceptance rate, the proposed method achieves improved performance. When all passengers accept the recommendations, the proposed method attains optimal robotaxi dispatch solely through NSGA-MC. Ideally, all passengers would accept the recommendations. To continuously approach this optimum, further exploration into the balance between pricing and cost may be necessary to incentivize higher recommendation acceptance rates.

5.8. Management insights

To improve the efficiency of robotaxi dispatch, this work proposes a two-stage method with the recommendation of PUDO points and BT, providing a novel plan for the robotaxi service and the travel of passengers. The new method involves changes to PUDO and BT, further optimizing routes of robotaxis and the robotaxi-request matching. The management insights are summarized as follows:

- 1) This study highlights actionable management strategies that contribute to the adoption and commercialization of robotaxis. The proposed method is also adaptable to HDTs where drivers adhere to dispatch arrangements. With the development of ride-hailing apps such as Uber, Lyft, and DiDi, this method is well-suited for integration into diverse platforms. Its adoption is foreseeable as robotaxi services scale up for commercialization.

- 2) The government and company can create some incentives to promote the application of this method, thereby improving the robotaxi utilization and traffic conditions. By encouraging passengers to accept the recommended PUDO points and BT through discounts, this method effectively reduces the detour, particularly in regions where lanes enforce directional travel. The experimental results demonstrate that with the increase in the request density, the average travel expense of passengers tends to decrease, and the profit per kilometer of robotaxis correspondingly increases.
- 3) The sensitivity analysis spans the entire range of acceptance rates, from 0% to 100%, which reveals that if more passengers accept the recommendation, the method is more effective. However, the profit declines slightly with the increase in acceptance rates, which means that more reasonable pricing policies should be developed further. Operators can pilot robotaxi services using this method and leverage the collected data to refine pricing policies, maximizing profit while attracting a larger customer base.

6. Conclusion

This work formulates mathematical models for a two-stage robotaxi dispatch problem that considers the variability of PUDO points and BT, along with passenger willingness, which is identified as an NP-hard problem. To provide specific recommendations of PUDO points and BT, the models include two stages: 1) the PUDO points and BT offsets are optimized; and 2) passengers decide whether to accept these specific recommendations via their willingness, and the robotaxi-request matching and the route of robotaxis are optimized. Then, we propose NSGA-MC and NSGA-IE to gain high-quality solutions and achieve two optimization objectives: minimizing the total travel expense of passengers and maximizing the profit per kilometer of robotaxis. Experimental results indicate that the proposed algorithms outperform alternatives in solving the formulated problem. Furthermore, the percentage of requests with a negative detour rate increases when more passengers accept recommendations, substantiating the performance of our method in reducing the travel distance of robotaxis. Consequently, the method not only reduces the travel expense of passengers to encourage them to adopt this transportation mode, but also generates positive profit for robotaxis by shortening travel distance. The novel method integrates carpooling with the recommendation of PUDO points and BT, enhancing the flexibility and diversity of robotaxi dispatch. It provides a valuable theoretical framework for addressing similar problems.

Given the developmental stage of robotaxis, this work has certain limitations. In the experiments, passenger willingness is randomly assigned, whereas in practice, it should be predictable through extensive data analysis. Pricing policies, influenced by many factors such as operating cost, city-specific economic conditions, and regulatory policies, also require deeper investigation. While this study explores the impact of the proposed method on the profit of robotaxis and the travel expense of passengers under a discounted-based pricing policy, a more comprehensive pricing policy remains to be developed. Furthermore, this work primarily addresses the allocation of reserved requests. The allocation of immediate requests, which involves greater complexity, has not been realized in the current framework and warrants future exploration, potentially using deep learning techniques. Additionally, supply-demand imbalances exist between robotaxi and requests at the city level. Most robotaxis are deployed in specific areas rather than widely used for long-distance transportation, making data acquisition and real-world testing difficult.

Future research will aim to address the identified limitations and explore several critical areas. The pricing policy has a significant impact on objective functions. Improving it is conducive to further reducing the travel expenses of requests and increasing the profit of robotaxis. To accurately capture passenger willingness, future research will focus on analyzing the factors that influence it. Once the trip data of robotaxis is available, we can forecast passenger willingness and then adopt a predict-then-optimize paradigm. Carpooling rules also affect the robotaxi dispatch. The impact of simultaneous carpooling of passengers in more than two requests on robotaxi dispatch requires further exploration. The dynamic appearance or cancellation of requests by passengers affects global dispatch. Additionally, real-time traffic is complex and uncertain, including traffic lights and traffic congestion. We will combine the optimization algorithm with reinforcement learning to deal with the aforementioned issues and solve a city-level robotaxi dispatch problem by developing a method to optimize the availability of robotaxis over time. Such city-level and large-scale dispatch will be key future work.

CRediT authorship contribution statement

Mengqi Li: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Liang Qi:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Formal analysis. **Wenjing Luan:** Writing – review & editing, Supervision. **Qurra Tul Ann Talukder:** Writing – review & editing. **Xiwang Guo:** Writing – review & editing. **Lu Liu:** Writing – review & editing, Validation, Methodology.

Declaration of competing interest

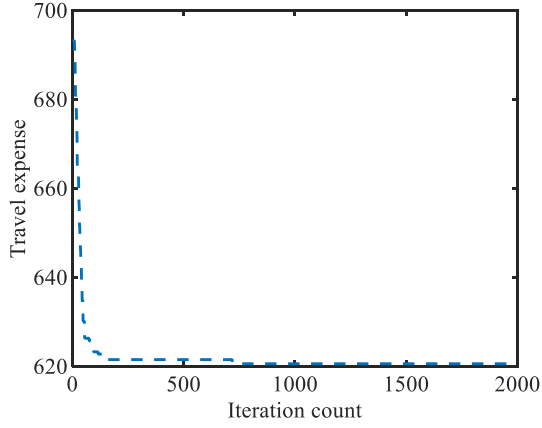
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

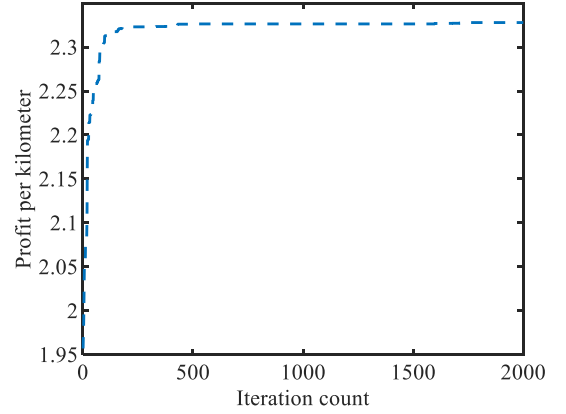
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Appendix A

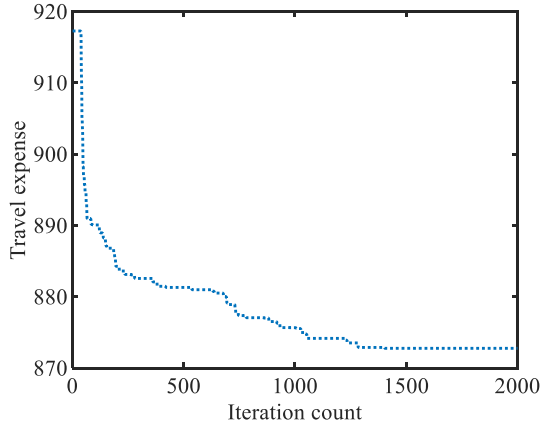
This appendix provides supplementary subplots for Figs. 11 and 13. Figs. A.1–A.2 show the results of the iterative process of the proposed algorithm and the impact of varying acceptance rates on recommendations on R_2 – R_4 , respectively.



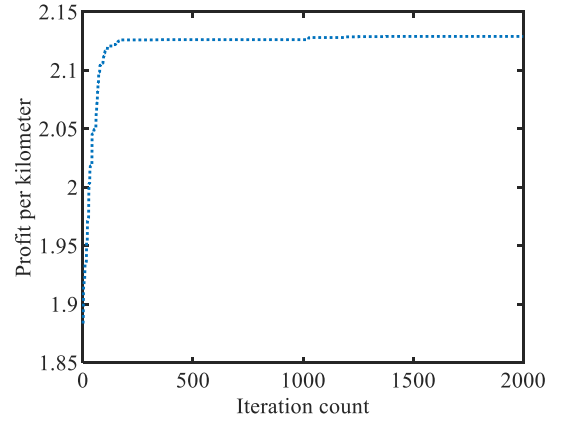
(a)



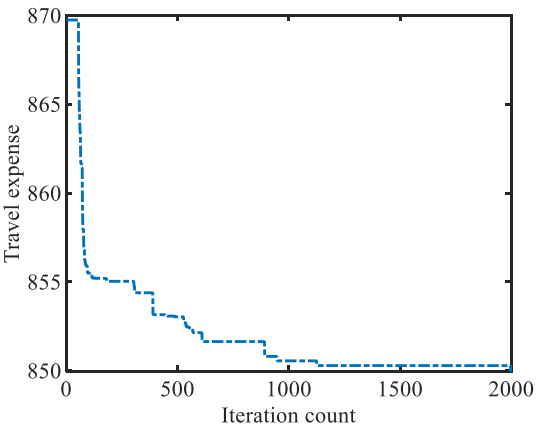
(b)



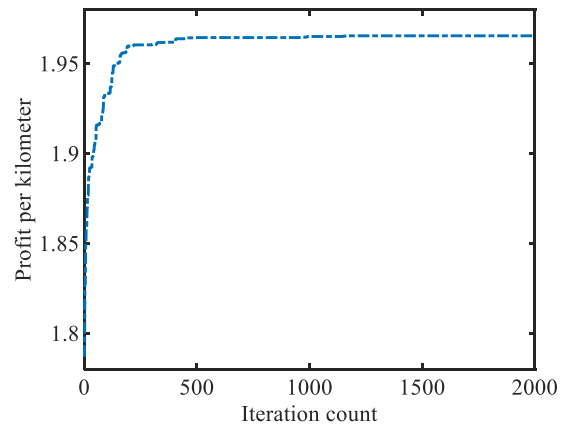
(c)



(d)



(e)



(f)

Fig. A1. Iterative process of the proposed algorithm: (a) Travel expense of R_2 ; (b) Profit per kilometer of R_2 ; (c) Travel expense of R_3 ; (d) Profit per kilometer of R_3 ; (e) Travel expense of R_4 ; and (f) Profit per kilometer of R_4 .

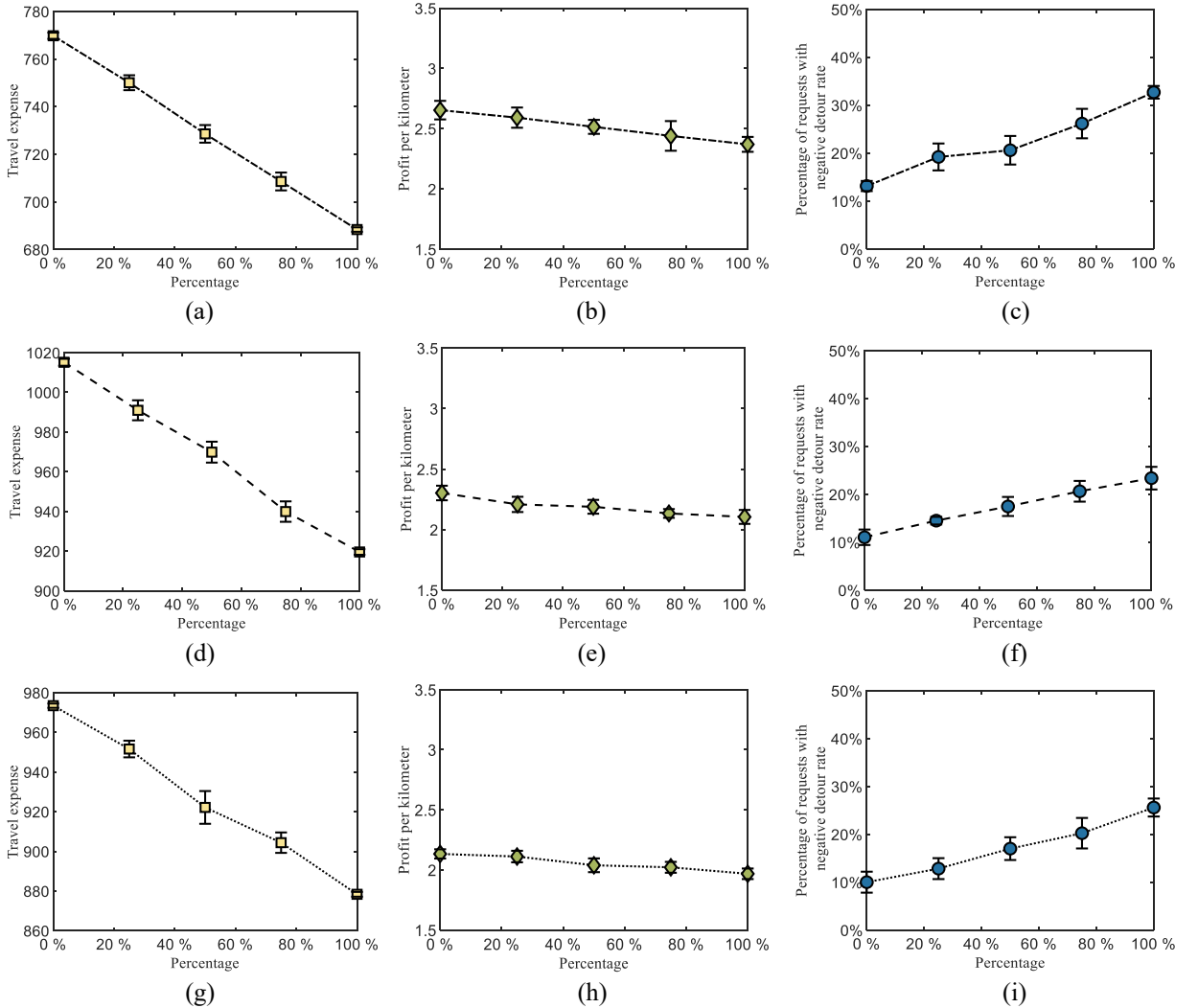


Fig. A2. The impact of varying acceptance rates on recommendation: (a) Travel expense of R_2 ; (b) Profit per kilometer of R_2 ; (c) Percentage of requests with a negative detour rate of R_2 ; (d) Travel expense of R_3 ; (e) Profit per kilometer of R_3 ; (f) Percentage of requests with a negative detour rate of R_3 ; (g) Travel expense of R_4 ; (h) Profit per kilometer of R_4 ; and (i) Percentage of requests with a negative detour rate of R_4 .

Data availability

Data will be made available on request.

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